

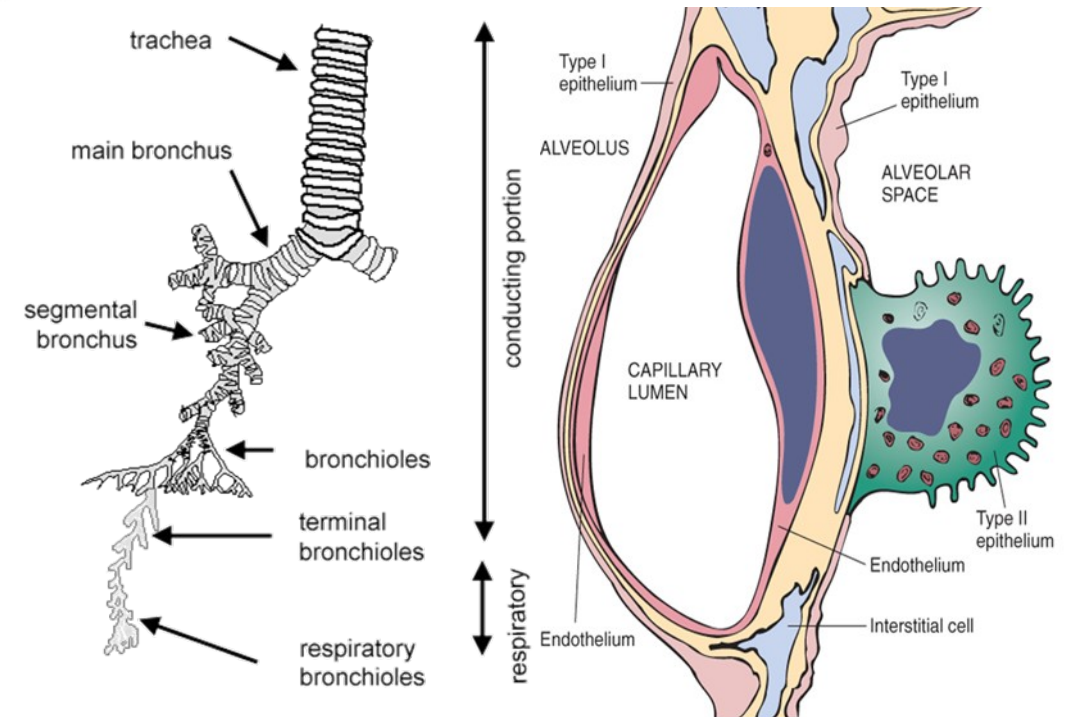
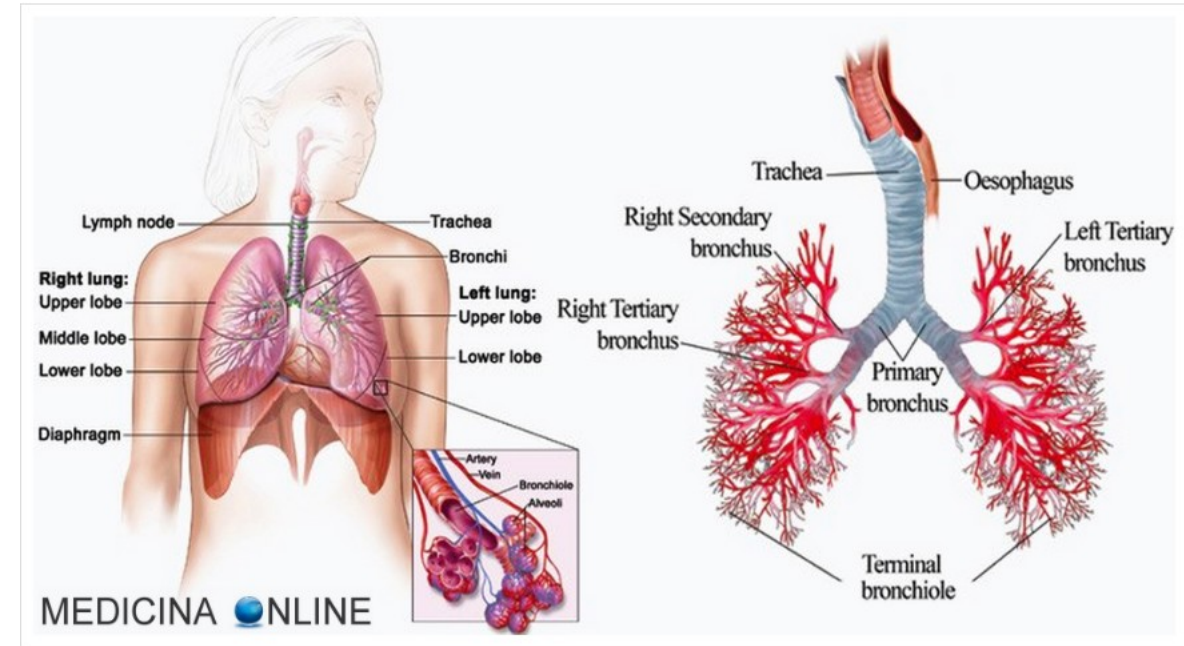
# **Diseases of the lower respiratory tract**

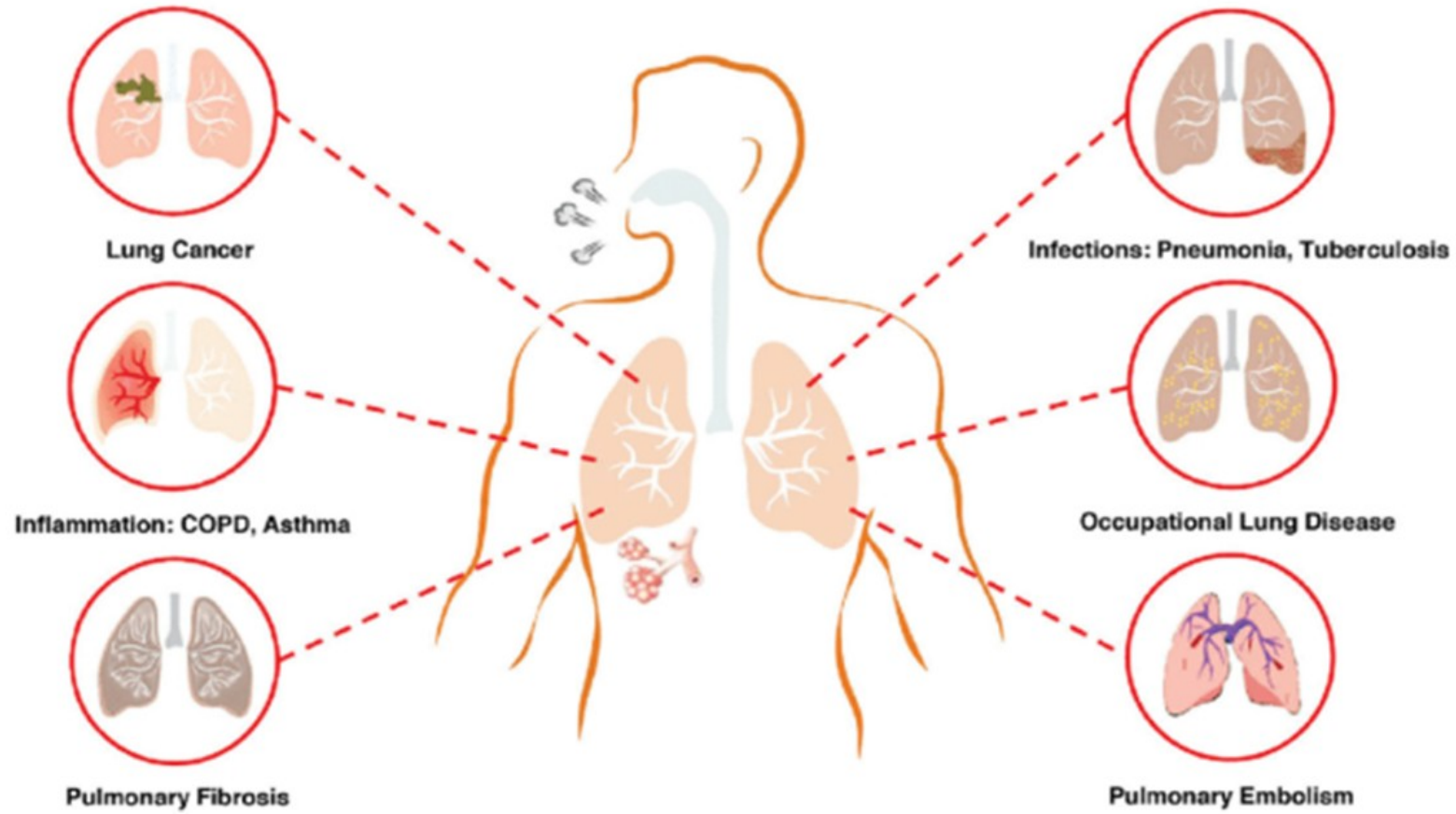
BY MELESE A.(MD, PATHOLOGIST)

# Introduction

## Anatomy:

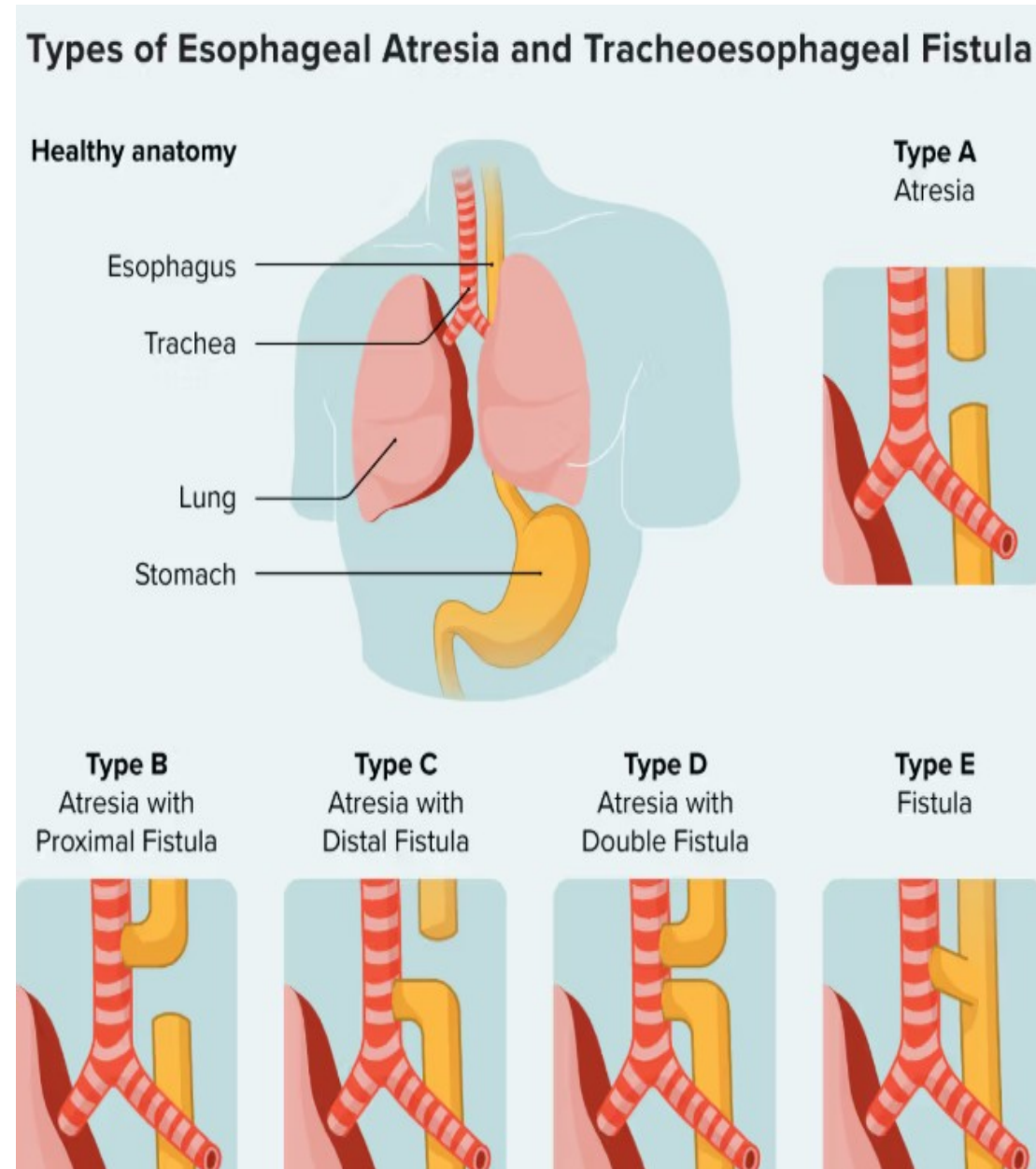
- ❖ The lower respiratory system consists of the:
  - Trachea
  - Bronchi and Bronchioles (the conducting airways)
  - Alveoli (the gas-exchange units)
  - Pleura (the lining of the lungs and chest cavity)
- ❖ **Bronchi** divide dichotomously to give **terminal** bronchioles → **Acinus**(respiratory bronchioles) → **alveolar ducts** → **alveolar sacs**(blind ends of respiratory passages)
- ❖ Function of the lung: **Oxygenation of blood** and **elimination of carbon dioxide**.





# Congenital Anomalies

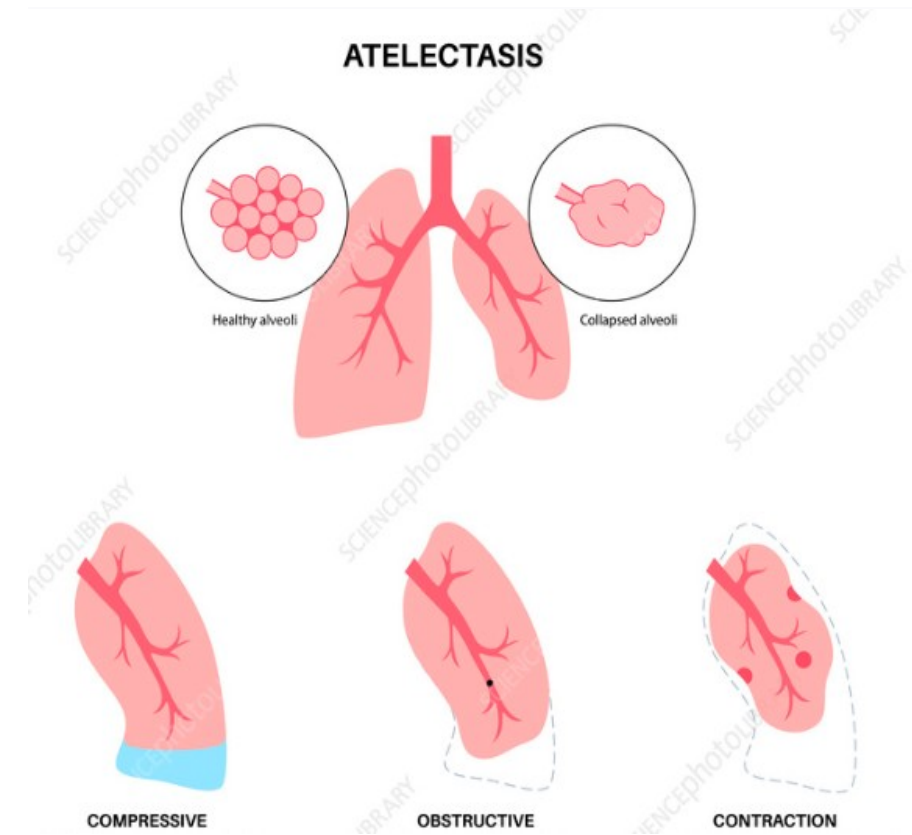
- ❖ Pulmonary hypoplasia
- ❖ Foregut cysts
- ❖ Pulmonary sequestration: a discrete area of lung tissue
  - lacks any connection to the airway system
  - has a **abnormal blood supply** arising from the **aorta** or its branches.
    - Extralobar sequestrations are **external** to the lung
    - Intralobar sequestrations occur **within** the lung.
- ❖ Tracheoesophageal Fistula (TEF) & Esophageal atresia
  - Anomalies involving an **abnormal connection (fistula)** between the **trachea and esophagus**, often with a **blind-ending pouch in the esophagus (atresia)**.
  - The most common type
    - **blind proximal esophagus** with a fistula from the **distal esophagus to the trachea**.





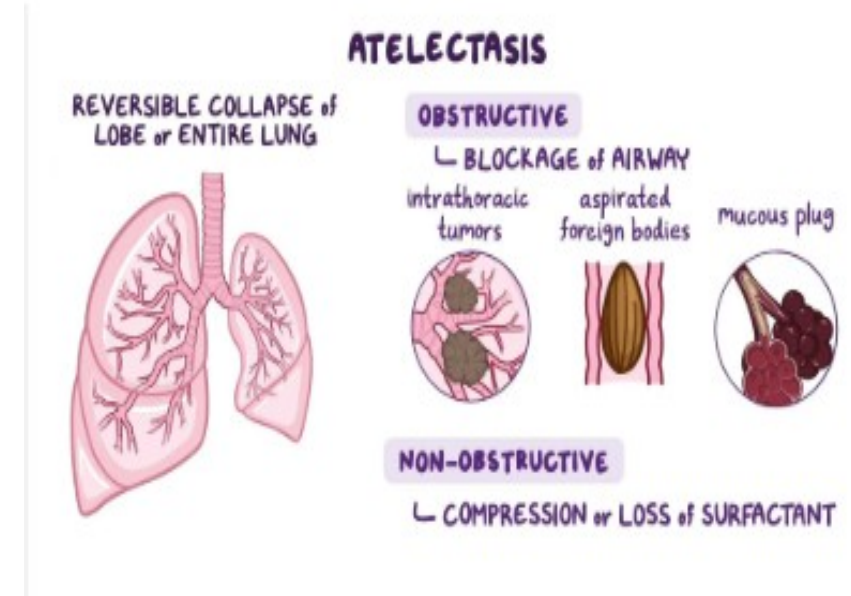
# Atelectasis (Collapse)

- ❖ Atelectasis refers either to
  - **Incomplete expansion** of the lungs (**neonatal** atelectasis)
  - **Collapse of previously inflated lung**, producing areas of relatively **airless** pulmonary parenchyma.
- ❖ Atelectasis is classified into **three forms**
  - **Resorption**
  - **Compression**
  - **Contraction**



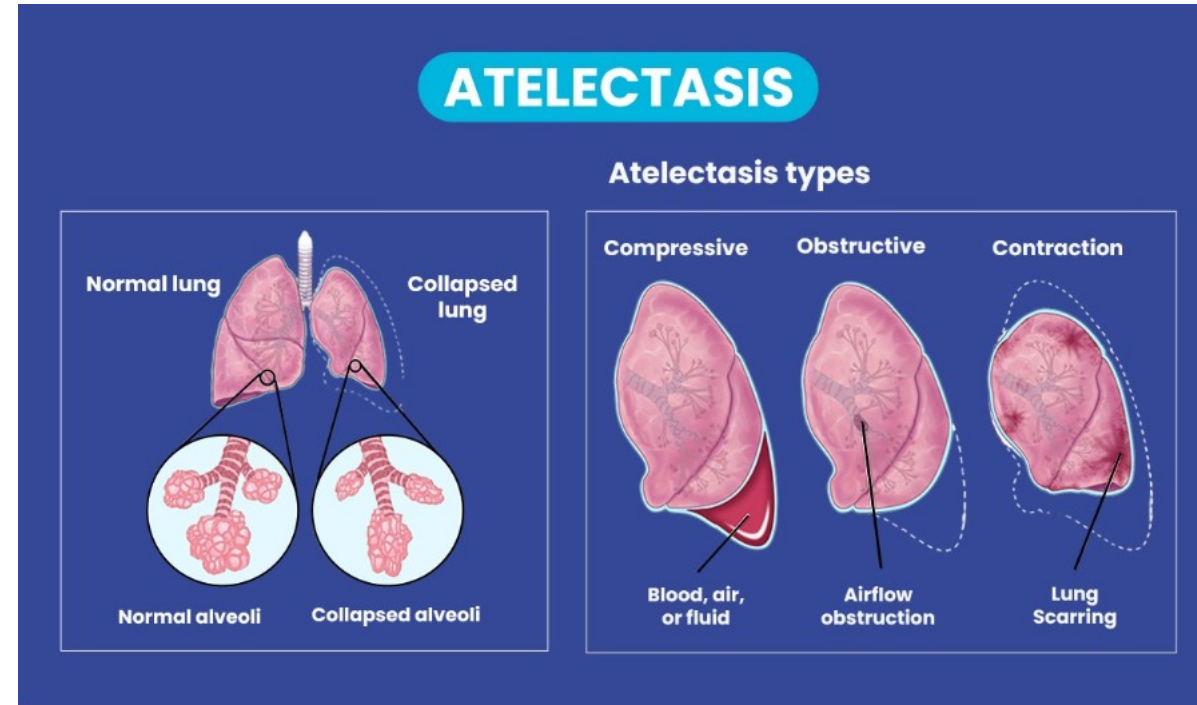
# Resorption Atelectasis

- ❖ Stems from **complete obstruction** of an airway.
- ❖ Occurs when an **obstruction prevents** air from reaching distal airways.
- ❖ The air already present gradually becomes **absorbed**, and **alveolar collapse** follows
- ❖ Depending on the **level of airway obstruction**, an **entire lung**, a **complete lobe**, or one or more **segments** may be involved
- ❖ Common cause is obstruction of a bronchus by
  - mucous or mucopurulent plug
  - Aspiration of foreign bodies
  - fragments of bronchial tumors
- ❖ Frequently occurs **postoperatively**
- ❖ The **mediastinum shifts toward** the atelectatic lung.



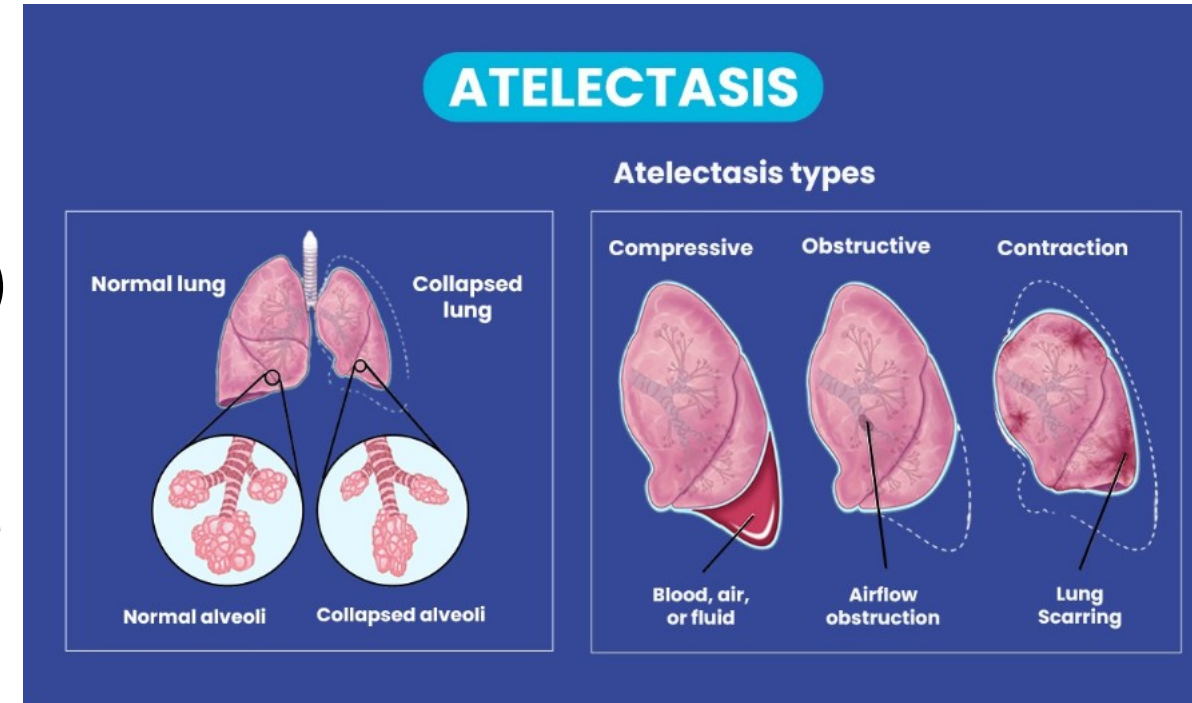
# Compression Atelectasis

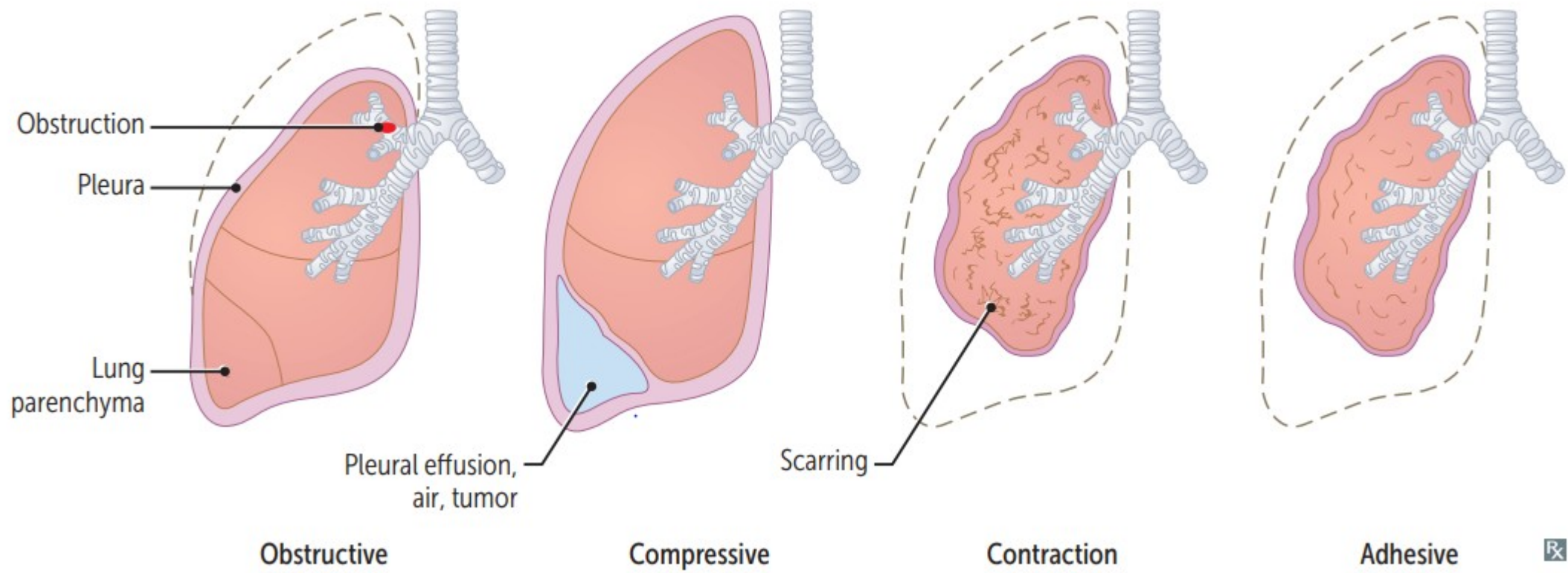
- ❖ Called **passive** or **relaxation** atelectasis
- ❖ Is usually associated with **accumulations of fluid, blood, or air** within the **pleural cavity**
- ❖ **Basal atelectasis** resulting from the **elevated position of the diaphragm** commonly occurs
  - In bedridden patients
  - in patients with ascites
  - in patients **during** and **after** surgery
- ❖ The **mediastinum** shifts ***away from*** the affected lung.



# Contraction Atelectasis

- ❖ Occurs when **focal** or **generalized** pulmonary or pleural **fibrosis** prevents **full lung expansion**.
- ❖ Atelectasis (except that caused by contraction) is potentially reversible
- ❖ Should be treated promptly to **prevent hypoxemia** and **superimposed infection** of the collapsed lung

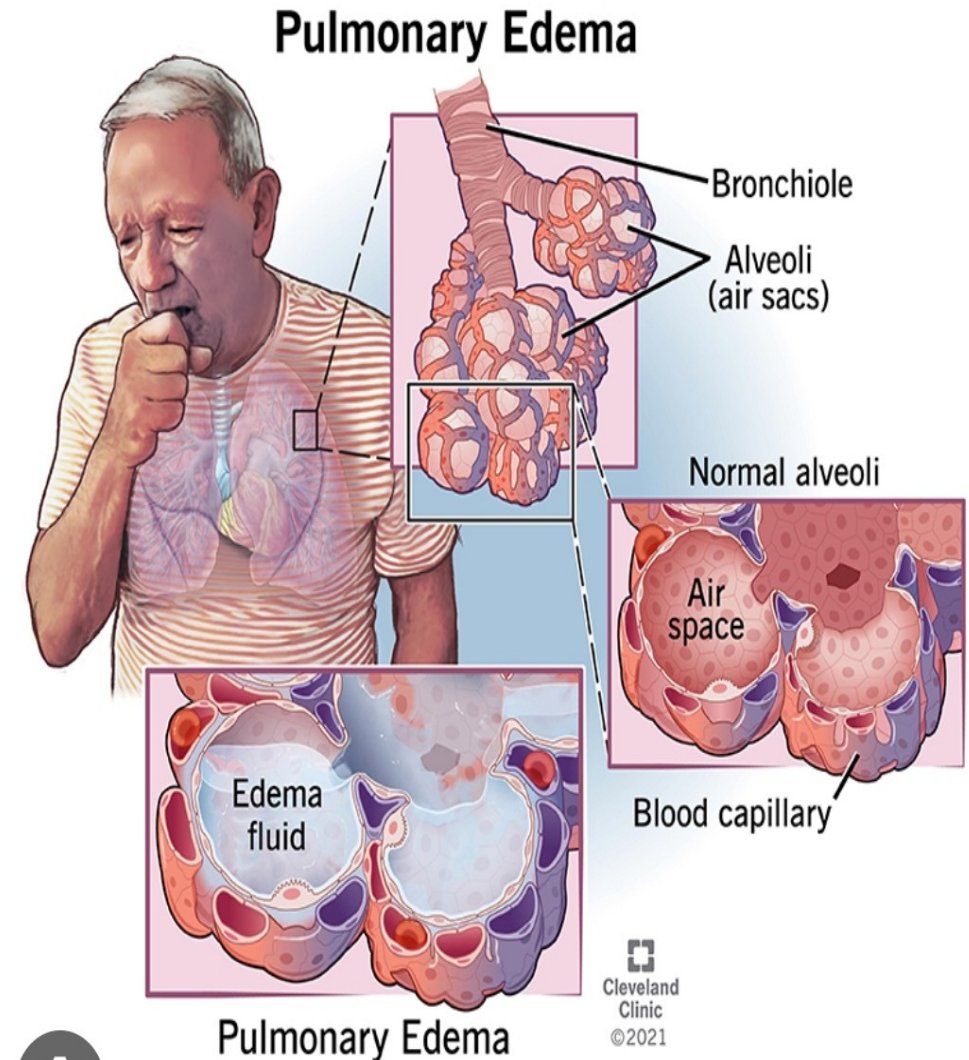






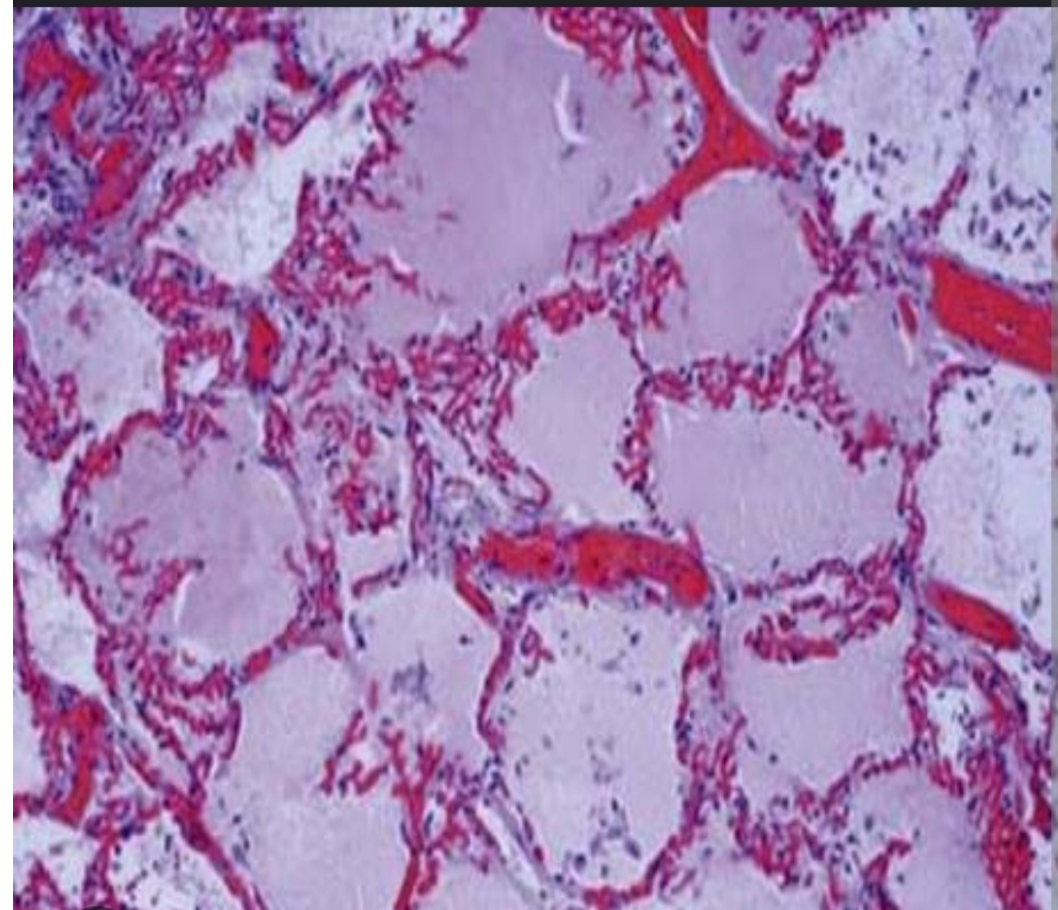
# Pulmonary Edema

- ❖ Abnormal accumulation of fluid in the lung parenchyma, primarily within the **alveoli** and **interstitial spaces**.
- ❖ Can result from
  - Hemodynamic disturbances (cardiogenic pulmonary edema)
  - Direct increases in capillary permeability (microvascular injury)



# Hemodynamic Pulmonary Edema

- ❖ Is due to **increased hydrostatic pressure**
  - occurs most commonly in **left sided congestive heart failure**.
- ❖ Fluid accumulates initially in the **basal regions** of the lower lobes because hydrostatic pressure is greatest in these sites (**dependent** edema).
- ❖ Histologically,
  - the alveolar **capillaries** are engorged
  - **intra-alveolar transudate** appears as ***finely granular pale pink*** material.
- ❖ Alveolar microhemorrhages and hemosiderin laden macrophages (“heart failure” cells) may be present.



# Edema Caused by Microvascular (Alveolar) Injury

- ❖ Non cardiogenic pulmonary edema is due to **injury to the alveolar septa**.
- ❖ Primary injury to the **vascular endothelium** or **damage to alveolar epithelial cells** (with secondary microvascular injury)

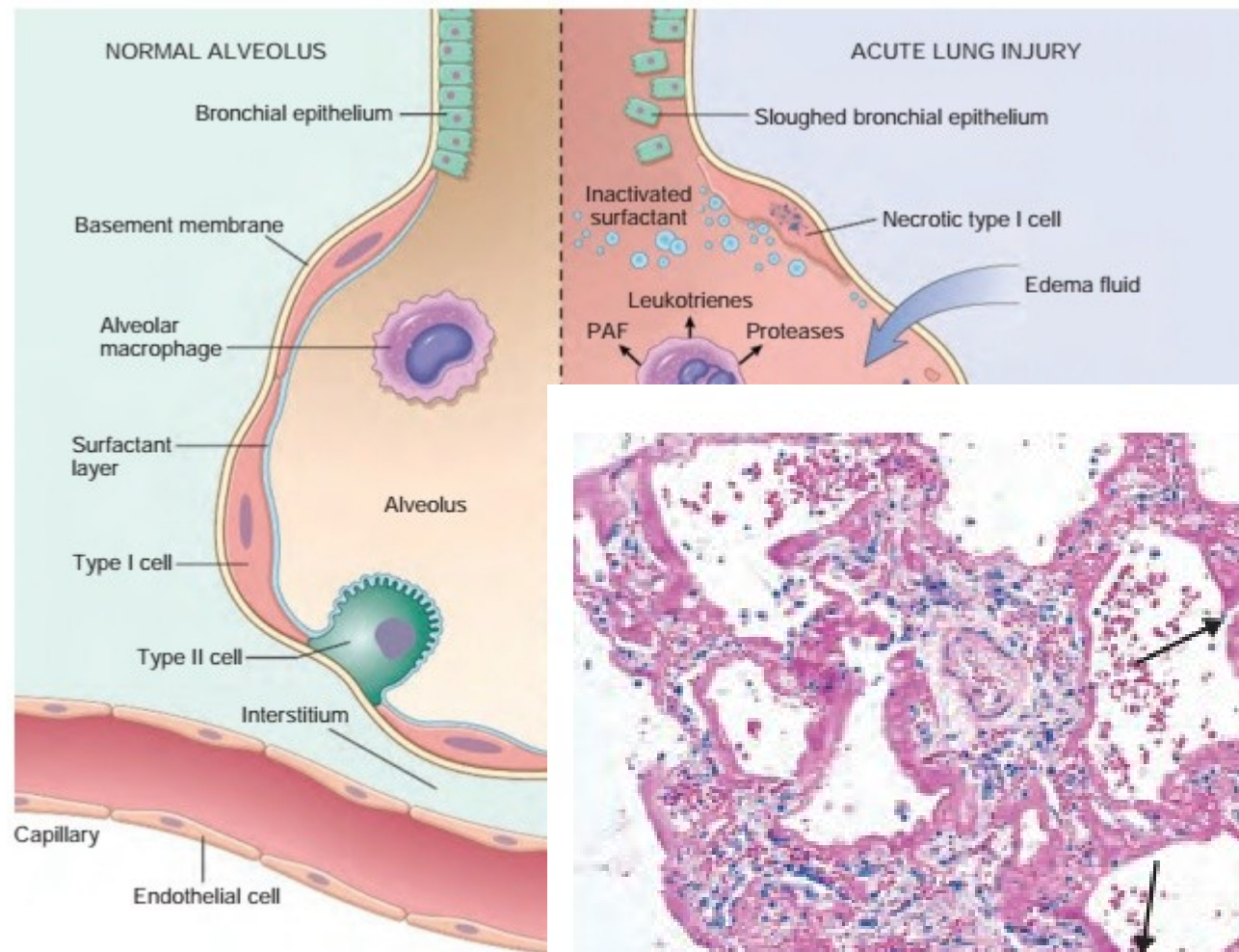
## Pathogenesis:

- **Direct Injury** (e.g., pneumonia, aspiration, smoke inhalation) or **Indirect Injury** (e.g., sepsis, trauma, pancreatitis) → Activation of inflammatory mediators (cytokines, neutrophils) → Damage to capillary endothelium and alveolar epithelium → Increased Permeability.
- ❖ Fluid, proteins, and even red blood cells leak freely into the alveolar spaces. The lymphatic system is overwhelmed.

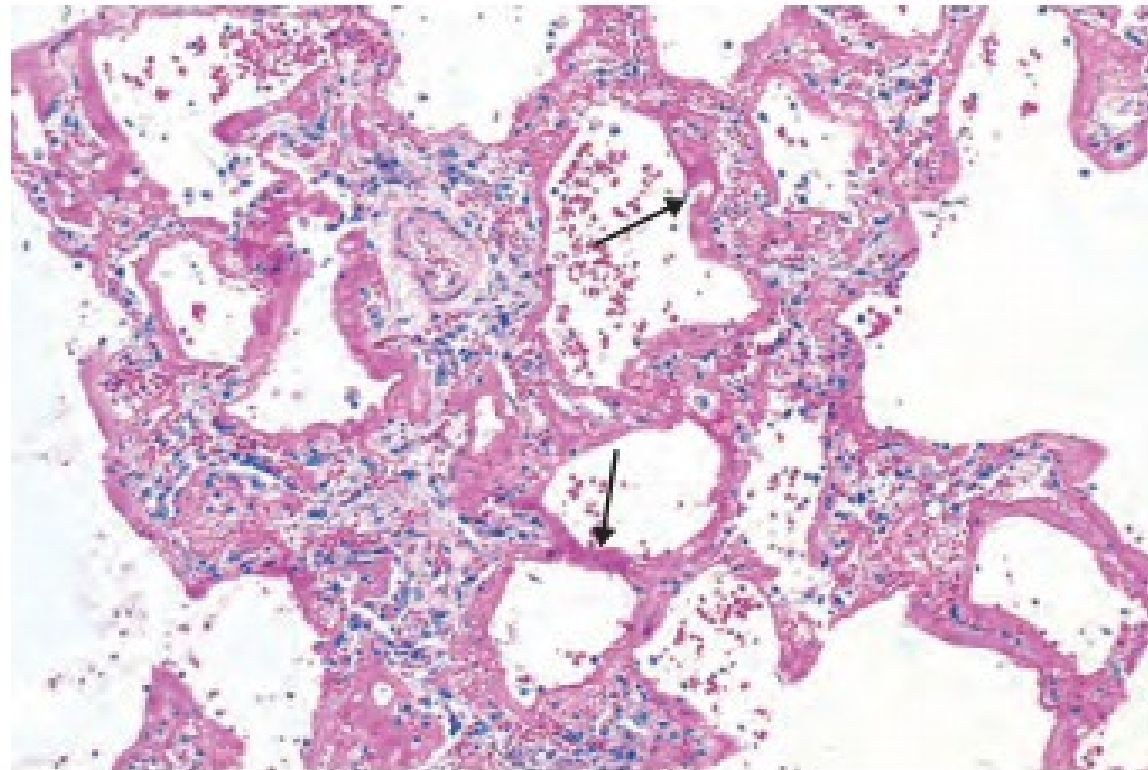
# Acute Lung Injury and Acute Respiratory Distress Syndrome (Diffuse Alveolar Damage)

- ❖ **Acute lung injury (ALI)** (also called **noncardiogenic pulmonary edema**)
  - Is characterized by the **abrupt onset** of *significant hypoxemia* and *bilateral pulmonary infiltrates* in the **absence of cardiac failure**.
- ❖ **Acute respiratory distress syndrome (ARDS)** is a manifestation of **severe ALI**.
- ❖ **ARDS** is a clinical syndrome of **progressive respiratory insufficiency** caused by **diffuse alveolar damage** in the setting of
  - sepsis, severe trauma, or diffuse pulmonary infection
- ❖ Both ARDS and ALI are associated with
  - **Increases in pulmonary vascular permeability, edema and epithelial cell death.**
- ❖ The **histologic** manifestation of these diseases is ***diffuse alveolar damage (DAD)***.
  - hyaline membranes (a characteristic feature of ALI/ARDS)





**Figure 15-3** The normal alveolus (*left side*) compared with the injured (Modified with permission from Matthay MA, Ware LB, Zimmerman leukin-1; *MIF*, migration inhibitory factor; *PAF*, platelet activating fact



**Figure 15-4** Diffuse alveolar damage (acute respiratory distress syndrome). Some of the alveoli are collapsed, while others are distended. Many are lined by hyaline membranes (*arrows*).



- ❖ Profound dyspnea & tachypnea herald ARDS
- ❖ Cyanosis, hypoxemia, resp.failure
- ❖ Hypoxemia becomes unresponsive to oxygen therapy
- ❖ Respiratory acidosis
- ❖ Mortality rate is about 60%
- ❖ X- ray may be normal initially, later there will be diffuse bilateral infiltrates

**Table 15.2 Conditions Associated With Development of Acute Respiratory Distress Syndrome**

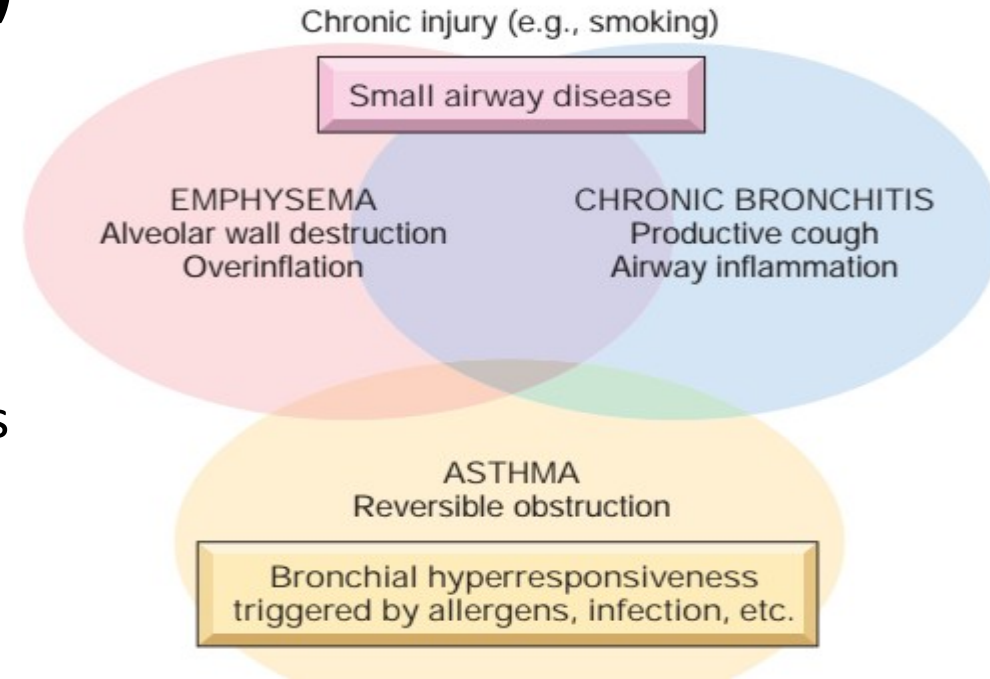
|                                                                                                                                                                                           |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Infection</b>                                                                                                                                                                          |
| Sepsis <sup>a</sup><br>Diffuse pulmonary infections <sup>a</sup><br>Viral, <i>Mycoplasma</i> , and <i>Pneumocystis</i> pneumonia; miliary tuberculosis<br>Gastric aspiration <sup>a</sup> |
| <b>Physical/Injury</b>                                                                                                                                                                    |
| Mechanical trauma including head injuries <sup>a</sup><br>Pulmonary contusions<br>Near-drowning<br>Fractures with fat embolism<br>Burns<br>Ionizing radiation                             |
| <b>Inhaled Irritants</b>                                                                                                                                                                  |
| Oxygen toxicity<br>Smoke<br>Irritant gases and chemicals                                                                                                                                  |
| <b>Chemical Injury</b>                                                                                                                                                                    |
| Heroin or methadone overdose<br>Acetylsalicylic acid<br>Barbiturate overdose<br>Paraquat                                                                                                  |
| <b>Hematologic Conditions</b>                                                                                                                                                             |
| Transfusion-associated lung injury (TRALI)<br>Disseminated intravascular coagulation                                                                                                      |
| <b>Pancreatitis</b>                                                                                                                                                                       |
| <b>Uremia</b>                                                                                                                                                                             |
| <b>Cardiopulmonary Bypass</b>                                                                                                                                                             |
| <b>Hypersensitivity Reactions</b>                                                                                                                                                         |
| Organic solvents<br>Drugs                                                                                                                                                                 |

<sup>a</sup>More than 50% of cases of acute respiratory distress syndrome are associated with these four conditions.

# Obstructive and Restrictive Lung Diseases

## Obstructive lung diseases (airway diseases)

- ❖ Are characterized by an **increase in resistance** to airflow due to **partial or complete obstruction** at any level
- ❖ Pulmonary function tests
  - **decreased** maximal airflow rates during forced **expiration** => expressed as **FEV1/FVC**
  - An FEV1/FVC ratio of **less than 0.7** generally indicates **airway obstruction**
- ❖ The major diffuse obstructive disorders
  - Emphysema
  - Chronic bronchitis
  - Bronchiectasis
  - Asthma

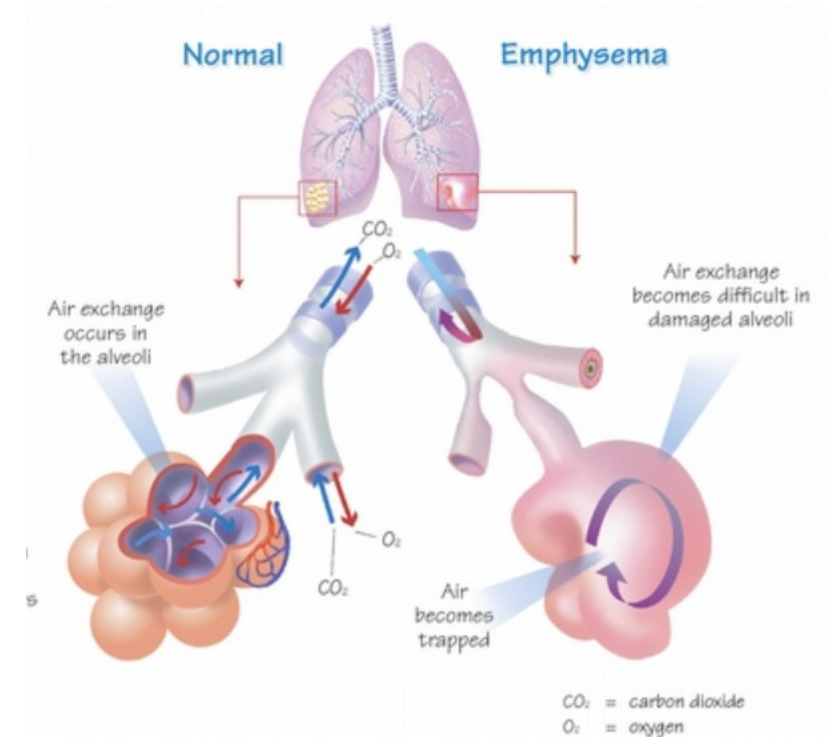


## Restrictive diseases

- ❖ Are characterized by **reduced expansion** of lung parenchyma and **decreased total lung capacity**.
- ❖ **Proportionate decreases** in both **total lung capacity** and **FEV1**, leading to **normal FEV1/FVC ratio**.
- ❖ FVC is reduced and the **expiratory flow rate** is **normal**
- ❖ Two broad kinds of conditions:
  - **chest wall disorders** (e.g., severe obesity, pleural diseases, kyphoscoliosis, and neuromuscular diseases such as poliomyelitis)
  - **chronic interstitial and infiltrative diseases**, such as pneumoconioses and interstitial fibrosis.

# Emphysema

- ❖ Is characterized by **irreversible enlargement** of the airspaces **distal to the terminal bronchiole**
  - accompanied by **destruction of their walls with subtle small air way fibrosis.**
- ❖ Emphysema is classified according to its ***anatomic distribution within the lobule***
- ❖ classified into four major types:
  - centriacinar
  - panacinar
  - paraseptal
  - irregular

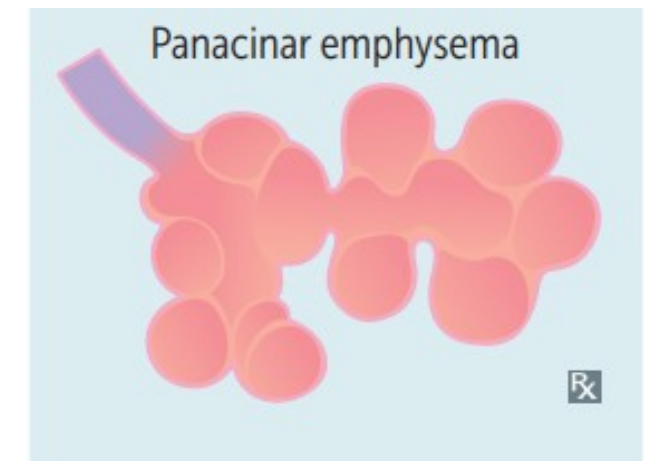
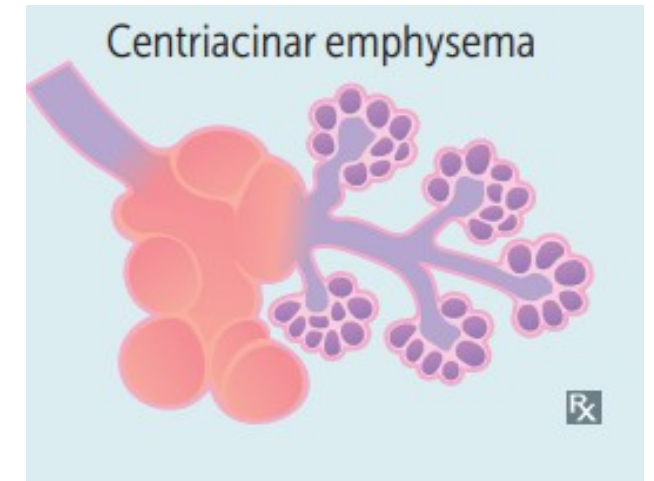


## 1. Centriacinar => The most common (>95%)

- Destruction of **central portion** with sparing of distal airways
- Both **emphysematous** and **normal** airspaces exist within the same acinus and lobule
- **Upper lobes** > lower
- Cause: **smoking**

## 2. Panacinar

- The acini are **uniformly enlarged**
- occur more commonly in the **lower zones** and in the anterior margins of the lung
- it is usually most severe at the **bases**
- is associated with ***α1-antitrypsin deficiency***





## Emphysema

### 3. Distal Acinar (paraseptal) Emphysema

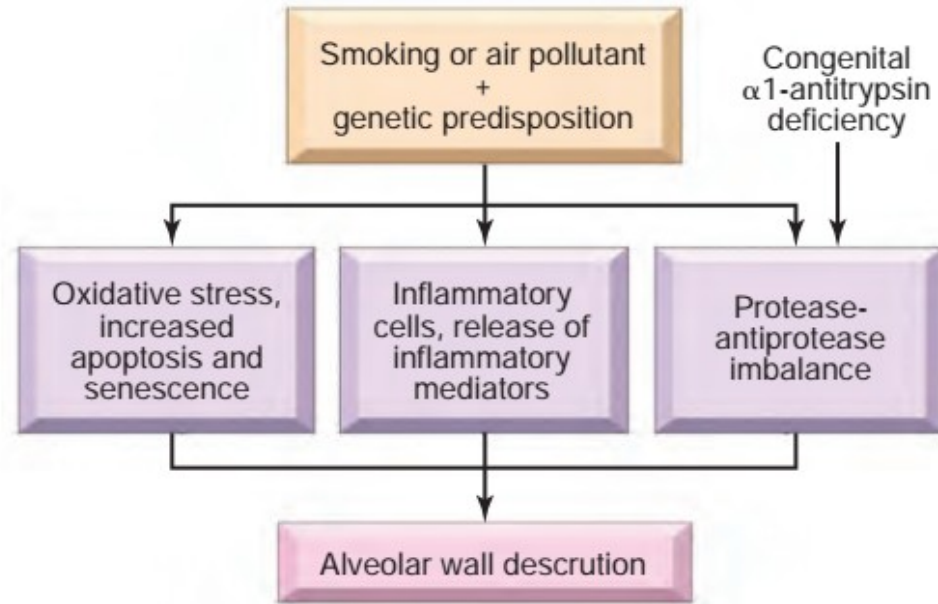
- ❖ The **proximal** portion of the acinus is normal, but the **distal** part is predominantly involved.
- ❖ More striking adjacent to the **pleura**
- ❖ Occurs adjacent to areas of **fibrosis, scarring** or **atelectasis**
- ❖ More **severe** in the **upper half** of the lungs
- ❖ Sometimes form cyst like structures
- ❖ Underlies many of the **spontaneous pneumothorax**

### 4. Airspace enlargement with fibrosis ( Irregular)

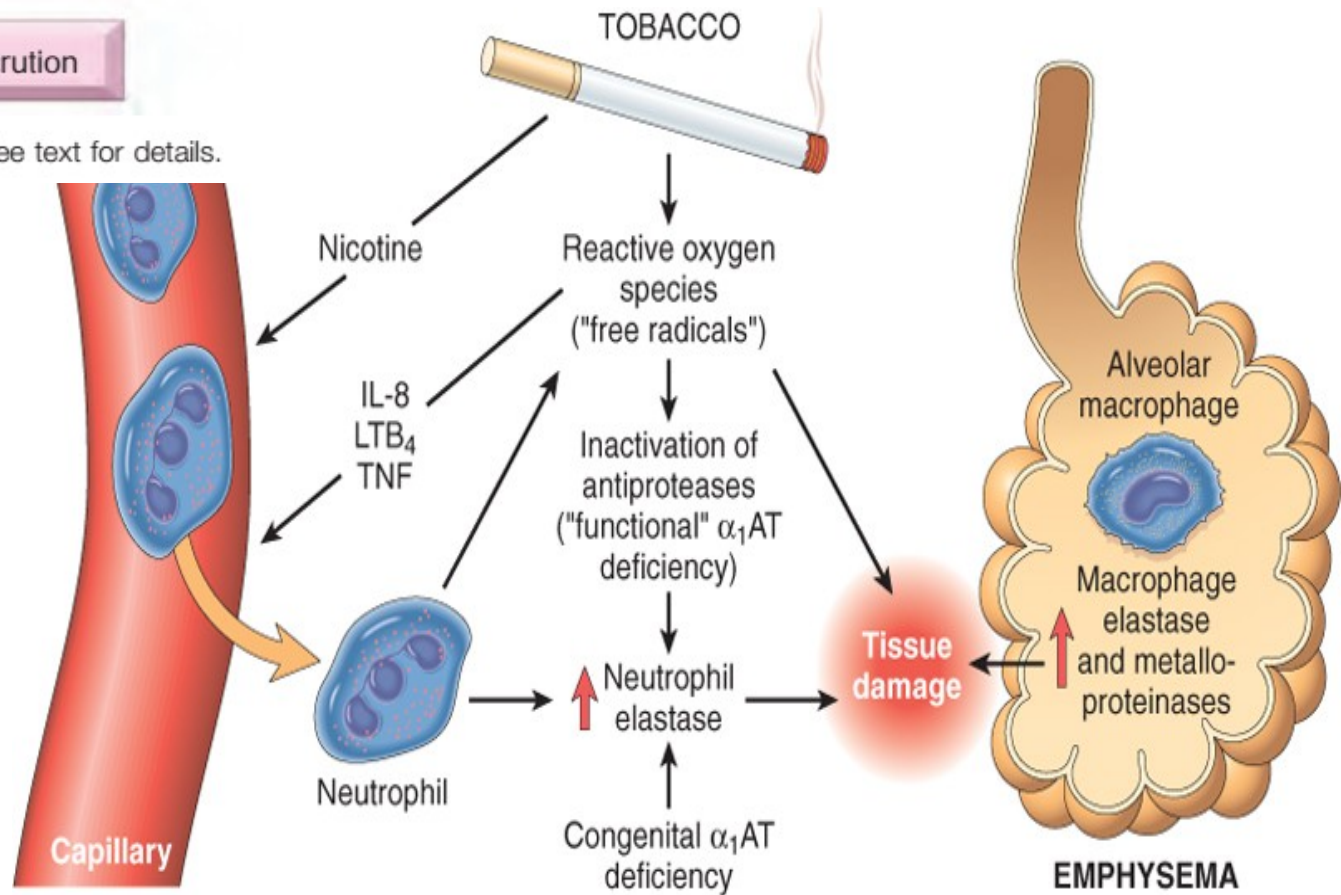
- ❖ The acinus is irregularly involved
- ❖ Associated with scarring
- ❖ Asymptomatic & clinically **insignificant**

# Pathogenesis

- ❖ Inhaled **cigarette smoke** and other noxious particles cause lung damage and inflammation, => which results in
  - **parenchymal destruction (emphysema)**
  - **airway disease** (bronchiolitis and chronic bronchitis)
- ❖ **Factors that influence the development of emphysema** include the following
  1. **Inflammatory mediators and leukocytes**
  2. **Protease-antiprotease imbalance.**
    - In **patients** who develop **emphysema**
      - There is a relative **deficiency of protective antiproteases**
  3. **Oxidant-antioxidant imbalance**
    - Substances in tobacco smoke, alveolar damage, and inflammatory cells all **produce oxidants**
    - results in **functional  $\alpha$ 1- AT deficiency**
  4. **Infection**



**Figure 15-8** Pathogenesis of emphysema. See text for details.



# MORPHOLOGY

- ❖ Advanced emphysema produces **voluminous lungs**
- ❖ Generally, the **upper two thirds** of the lungs are more severely affected.
- ❖ Large apical blebs or bullae are more characteristic of
  - **irregular emphysema** secondary to scarring
  - **distal acinar emphysema**.
- ❖ Microscopically
  - abnormally large alveoli are separated by thin septa with only focal centriacinar fibrosis.
- ❖ **Prolonged vasoconstriction** leads to changes of **pulmonary arterial HTN**.



Figure 15-9 Bullous emphysema. Note the large subpleural bullae (upper left).

# Clinical Course

- ❖ Symptoms do not appear **until** at **least one third** of the functioning pulmonary parenchyma is damaged.
- ❖ **Dyspnea** usually appears first, beginning **insidiously** but **progressing steadily**.
- ❖ In some patients, **cough** or **wheezing** is the chief complaint, easily confused with asthma.
- ❖ Cough and expectoration
- ❖ **Weight loss** is common and can be so severe as to suggest an **occult cancer**.
- ❖ Classically, the patient with **severe emphysema** is **barrel-chested** and **dyspneic**
  - **sits forward in a hunched-over** position, and breathes through **pursed lips**
- ❖ Such patients may **over ventilate** and remain well oxygenated (**pink puffers**)

## EMPHYSEMA

- “Barrel chest” from lung over-expansion
- Pursed-lip breathing
- Use of accessory muscles
- Tachypnoea
- Weak & Thin
- Decreased lung sounds





# Clinical Course

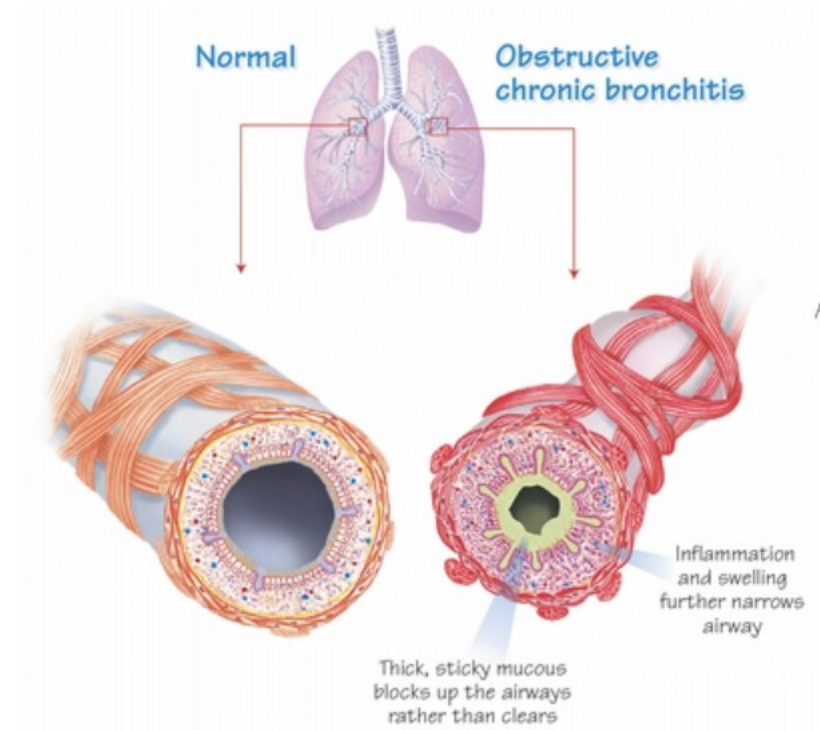
- ❖ Development of **cor pulmonale** and eventually **congestive heart failure**, related to **secondary pulmonary hypertension**, is associated with a **poor prognosis**.
- ❖ Death in most patients with emphysema is due to
  - coronary artery disease
  - respiratory failure
  - right sided heart failure
  - massive collapse of the lungs secondary to **pneumothorax**
- ❖ Treatment options include
  - smoking cessation
  - oxygen therapy
  - long-acting bronchodilators with inhaled corticosteroids

# Chronic Bronchitis

- ❖ Is defined clinically as
  - **persistent cough** with **sputum production** for **at least 3 months** in **at least 2 consecutive years**, in the **absence** of any other identifiable **cause**.
- ❖ Common in **habitual smokers** and inhabitants of **smog-laden cities**
- ❖ When chronic bronchitis persists for years, lead to
  - Cor pulmonale and heart failure
  - Cause metaplasia and dysplasia => providing a **rich soil** for **cancerous transformation**.

# Pathogenesis

- ❖ The **primary or initiating factor** in the genesis of chronic bronchitis is exposure to
  - **tobacco smoke (90% )**
  - **dust** from grain, cotton, and silica.
- ❖ **Mucus hypersecretion**
  - earliest feature
  - associated with hypertrophy of the submucosal glands
- ❖ **Inflammation**
  - Long-standing inflammation and accompanying **fibrosis** can also lead to **chronic airway obstruction**.
- ❖ **Infection**
  - significant in maintaining it and may be critical in producing **acute exacerbations**.

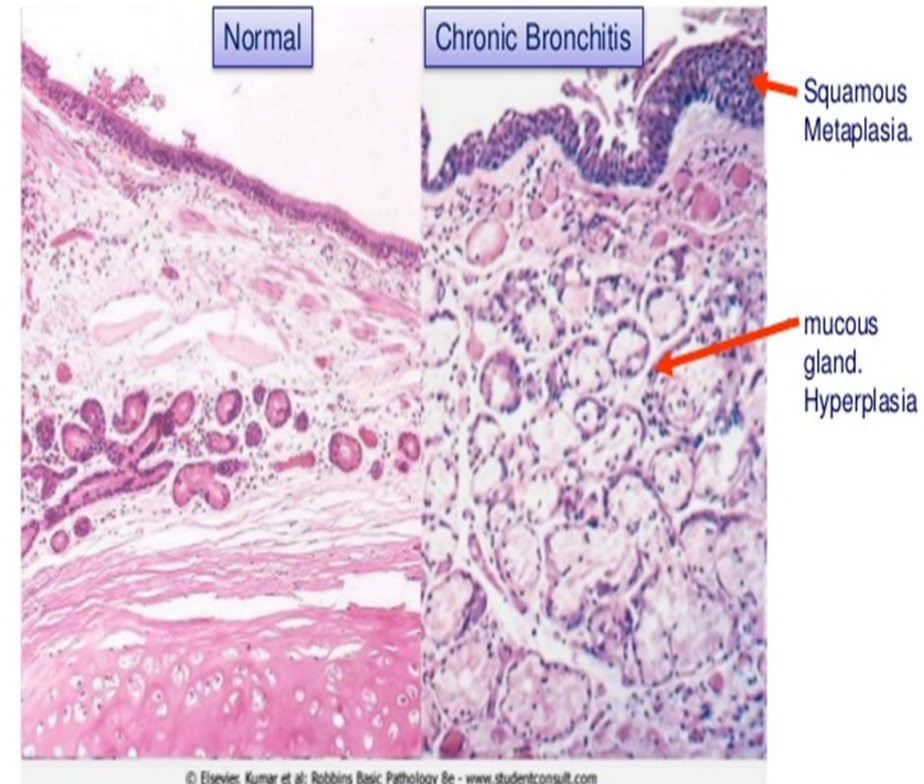


# MORPHOLOGY

- ❖ Grossly, there is **hyperemia**, **swelling**, and **edema** of the **mucous membranes**
- ❖ The characteristic features are
  - **Mild chronic inflammation** of the airways
  - Enlargement of the **mucus-secreting glands**
    - Assessed by the **ratio** of the **thickness** of the **mucous gland layer** to the **thickness of the wall** between the epithelium and the cartilage (**Reid index**).
    - The Reid index (normally 0.4) is **increased** in chronic bronchitis
- ❖ The bronchial epithelium may exhibit **squamous metaplasia and dysplasia**.
- ❖ There is **marked narrowing of bronchioles** caused by
  - ❖ mucus plugging, inflammation, and fibrosis.
- ❖ In the most **severe cases**, there may be **obliteration** of lumen **due to fibrosis (bronchiolitis obliterans)**.

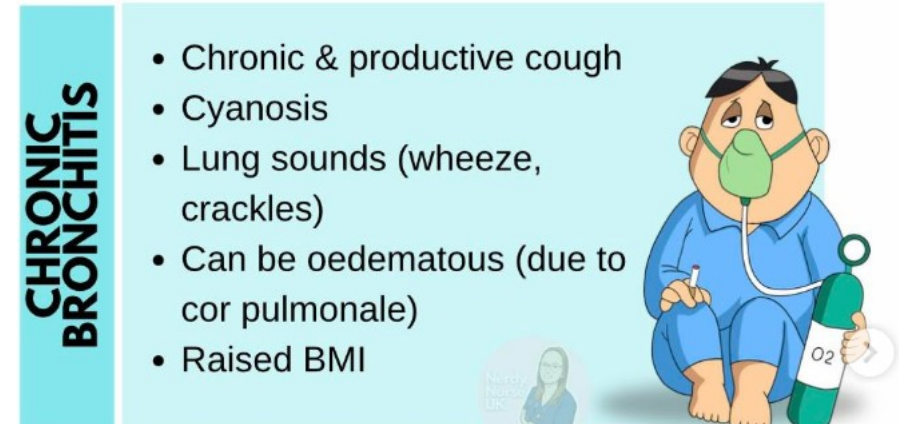


## Chronic Bronchitis:



# Clinical Features

- ❖ The cardinal symptom of chronic bronchitis is a **persistent cough productive of sputum**
- ❖ For many years **no other** respiratory functional impairment is present
- ❖ But eventually **dyspnea on exertion** develops.
- ❖ With the **passage of time**, and usually with **continued smoking**,
  - hypercapnia, hypoxemia, and mild **cyanosis** (“**blue bloaters**”).
- ❖ **Long-standing severe** chronic bronchitis commonly leads to
  - **cor pulmonale and cardiac failure.**
- ❖ Death may also result from
  - ❖ impairment of respiratory function due to **superimposed acute infections.**



**Table 15-4 Emphysema and Chronic Bronchitis**

|                           | Predominant<br>Bronchitis         | Predominant<br>Emphysema        |
|---------------------------|-----------------------------------|---------------------------------|
| Age (yr)                  | 40-45                             | 50-75                           |
| Dyspnea                   | Mild; late                        | Severe; early                   |
| Cough                     | Early; copious sputum             | Late; scanty sputum             |
| Infections                | Common                            | Occasional                      |
| Respiratory insufficiency | Repeated                          | Terminal                        |
| Cor pulmonale             | Common                            | Rare; terminal                  |
| Airway resistance         | Increased                         | Normal or slightly<br>increased |
| Elastic recoil            | Normal                            | Low                             |
| Chest radiograph          | Prominent vessels;<br>large heart | Hyperinflation; small<br>heart  |
| Appearance                | Blue bloater                      | Pink puffer                     |



# Asthma

- ❖ Asthma is a **chronic disorder** of the **conducting airways**
- ❖ Usually caused by an **immunological reaction**
- ❖ Is marked by **episodic bronchoconstriction** due to **increased airway sensitivity** to a variety of stimuli
- ❖ inflammation of the bronchial **walls** and **increased mucus secretion**.
- ❖ The disease is manifested by
  - Recurrent episodes of wheezing
  - breathlessness, chest tightness
  - cough, particularly at **night** and/or in the **early morning**.

# 1. Depending on the **frequency & severity** of symptoms

Mild

Moderate

Severe

- Rarely, a state of **unremitting attacks**, called **acute severe asthma**

# 2. Inciting agent

Intrinsic (non atopic)

Extrinsic (atopic)

In either type, episodes of bronchospasm can have **diverse triggers**, such as

**respiratory infections** (especially viral infections),

**exposure to irritants** (e.g., smoke, fumes), cold air, stress, and exercise.

Asthma may also be **classified** according to the **agents** or **events** that trigger bronchoconstriction. Include

- ✓ seasonal
- ✓ exercise-induced
- ✓ drug-induced (e.g., aspirin), and
- ✓ occupational asthma, and
- ✓ asthmatic bronchitis in smokers.

- ❖ Classic example of **IgE-mediated (type I) hypersensitivity** reaction
- ❖ Usually begins in **childhood**
- ❖ Triggered by **env'tal pollutants**
- ❖ Positive **family history** is common
- ❖ There is evidence of **allergen sensitization** and **immune** activation
- ❖ Asthmatic attacks often **preceded** by **allergic rhinitis, urticaria** or **eczema**
- ❖ Skin test is **positive** to allergen injection

# Pathogenesis

- ❖ The contributions of the **immune response**, **genetics** and **environment** are closely intertwined.
- ❖ **TH2 Responses, IgE and Inflammation.**
- ❖ A fundamental abnormality in asthma is
  - an **exaggerated TH2** response to normally **harmless environmental antigens**
- ❖ TH2 cells secrete **cytokines** that promote inflammation and stimulate **B cells** to produce **IgE** and other **antibodies**.
- ❖ These cytokines include
  - **IL-4**, which stimulates the production of **IgE**
  - **IL-5**, which activates locally recruited **eosinophils**
  - **IL-13**, which stimulates **mucus secretion** and also **promotes IgE production by B cells**
- ❖ **IgE** binds to the **Fc receptors** on submucosal **mast cells**
  - **repeat exposure** to the allergen triggers the mast cells to release granule contents and produce cytokines and other mediators
  - **collectively** induce the **early-phase (immediate hypersensitivity) reaction** and the **late phase reaction**.

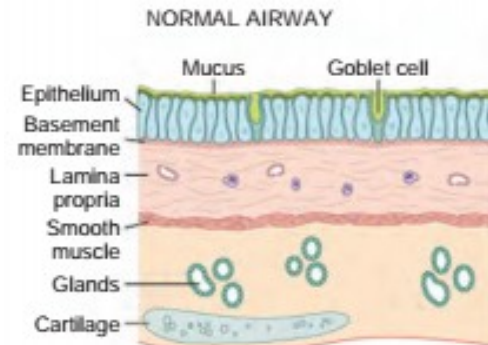
# Pathogenesis

- ❖ The **early reaction** is
  - dominated by **bronchoconstriction**
  - increased **mucus production**
  - variable degrees of **vasodilation**
  - increased **vascular permeability**
- ❖ **Bronchoconstriction** is **triggered** by **direct stimulation of subepithelial vagal** (parasympathetic) receptors.
- ❖ The late-phase reaction is **dominated** by recruitment of **leukocytes**, notably **eosinophils, neutrophils**, and more **T cells**.

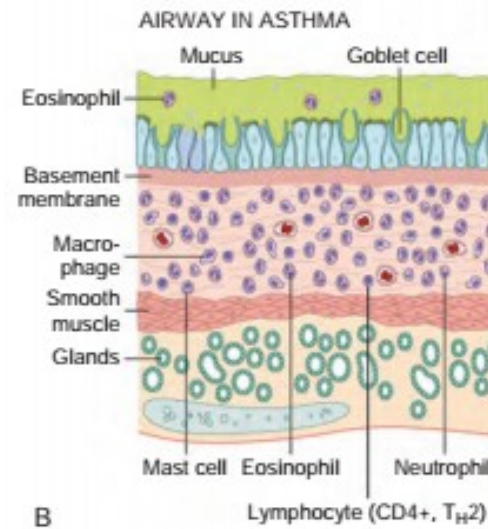
# Mediators

- ❖ **Leukotrienes C4, D4, and E4**, cause
  - prolonged bronchoconstriction
  - increased vascular permeability and
  - increased mucus secretion
- ❖ **Acetylcholine**: can cause airway **smooth muscle constriction** by directly **stimulating** muscarinic receptors
- ❖ **Histamine**: a potent bronchoconstrictor
- ❖ **Prostaglandin D2**: elicits bronchoconstriction and vasodilatation
- ❖ **Platelet-activating factor**: causes aggregation of platelets and release of serotonin from their granules

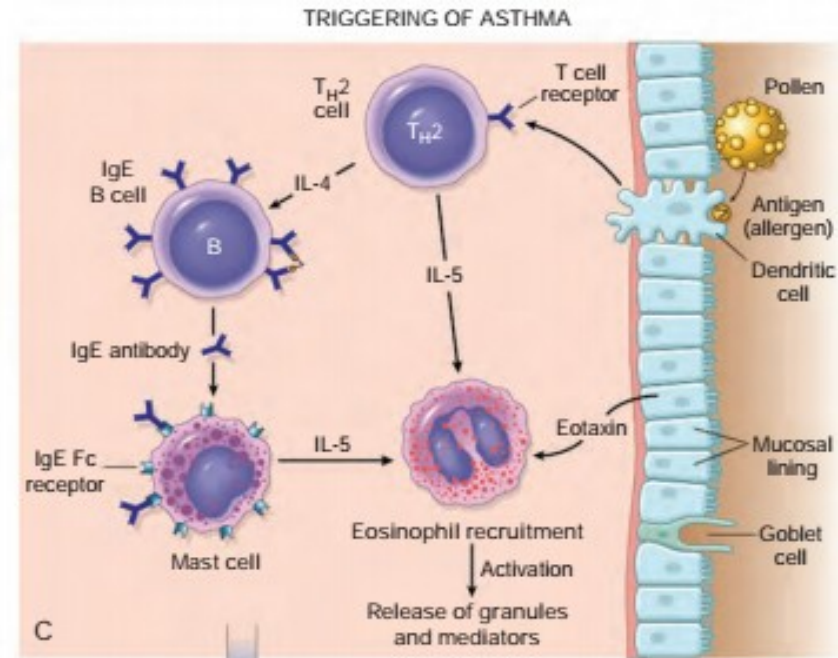




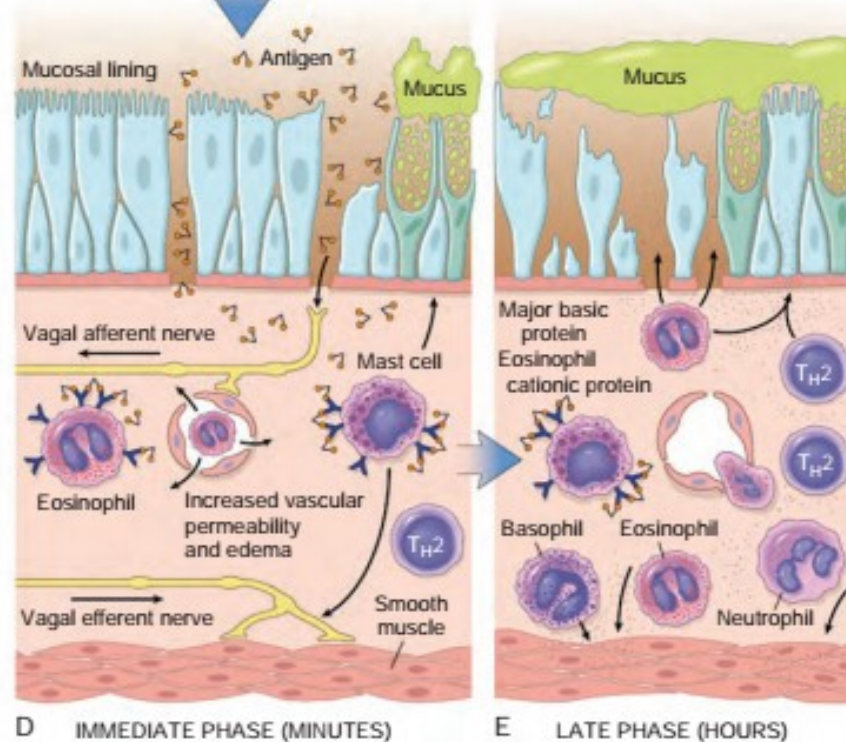
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B



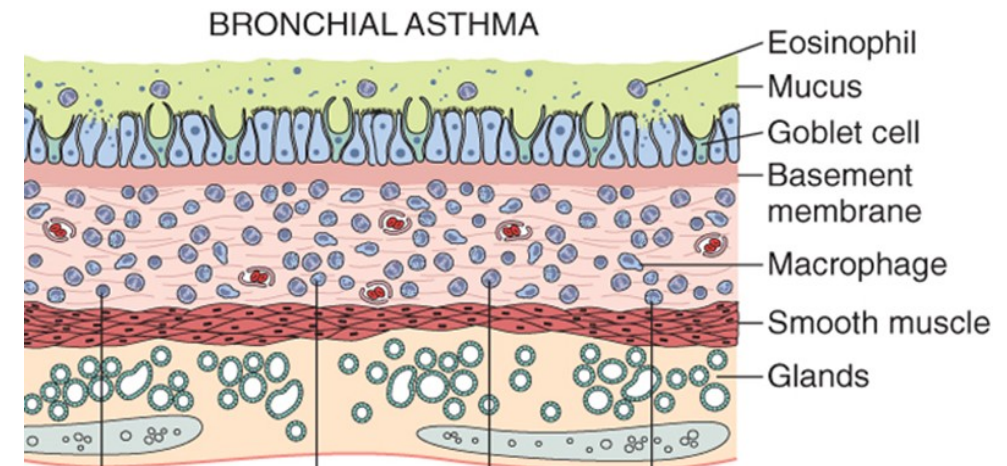
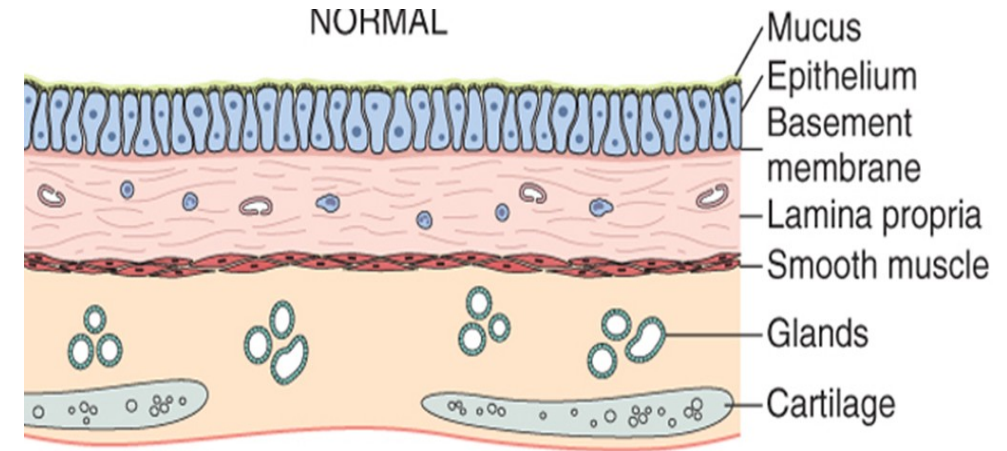
C



# MORPHOLOGY

- ❖ In patients dying of acute **severe asthma** (status asthmaticus) the lungs
  - Are **distended by overinflation** and
  - contain small areas of **atelectasis**.
- ❖ The most striking gross finding is **occlusion** of bronchi and bronchioles by
  - **thick, tenacious mucus plugs**, which often contain **shed epithelium**.
- ❖ A characteristic finding in **sputum** or **bronchoalveolar lavage** specimens is **Curschmann spirals**
  - result from **extrusion of mucus plugs** from subepithelial mucous gland ducts or bronchioles.
- ❖ Also present are numerous eosinophils
  - **Charcot-Leyden crystals**; are composed of an eosinophil protein called **galectin-10**.

- ❖ The other characteristic histologic findings (“**airway remodeling**”) include:
- Thickening of airway wall
  - Sub-basement membrane fibrosis (due to deposition of type I and III collagen)
  - Increased vascularity
  - An increase in the size of the **submucosal glands** and number of airway goblet cells
  - Hypertrophy and/or hyperplasia of the bronchial wall **muscle**



# Clinical Course

- ❖ The cardinal symptoms
  - chest tightness
  - dyspnea
  - wheezing, and coughing (with or without sputum production)
- ❖ In its most **severe form, status asthmaticus**
  - airflow obstruction that is so extreme that **marked cyanosis or even death ensues.**
- ❖ The diagnosis is based on demonstration of an **increase in airflow obstruction**
  - difficulty with exhalation (prolonged expiration, wheeze)
  - peripheral blood eosinophilia, and
  - the finding of **eosinophils, Curschmann spirals, and Charcot-Leyden crystals** in the sputum
- ❖ Up to **50% of childhood asthma remits in adolescence.**

- ❖ **Do not** have evidence of allergen sensitization
- ❖ Most frequently triggered by **viral** respiratory infection  
(e.g., rhinovirus, parainfluenza virus and respiratory syncytial virus)
- ❖ Positive family history is **uncommon**
- ❖ Serum **IgE** level is **normal**, **Skin test** is negative
- ❖ Virus induced inflammation of the respiratory mucosa lowers the threshold of the sub epithelial vagal receptors to irritants
- ❖ **Inhaled air pollutant** may also contribute

## Drug induced Asthma

- ❖ Aspirin – tipping the balance toward **bronchoconstrictor leukotrienes**
- ❖ Aspirin (and other NSAIDs) triggers asthma in these patients by **inhibiting the cyclooxygenase leading to a rapid decrease in prostaglandin E2.**
  - Normally prostaglandin E2 inhibits enzymes that generate proinflammatory mediators such as leukotrienes B4, C4, D4 and E4

**Occupational asthma-** This form of asthma may be triggered by fumes (epoxy resins, plastics), organic and chemical dusts (wood, cotton, platinum), gases (toluene), or other chemicals (formaldehyde, penicillin products)

❖ Is a disorder in which destruction of **smooth muscle** and **elastic tissue** by chronic necrotizing infections leads to

➤ **permanent dilation** of **bronchi** and **bronchioles**

❖ Manifested by cough ,fever, & copious amounts of foul smelling position dependent ,purulent sputum

❖ Because of better control of lung infections, bronchiectasis is now uncommon.

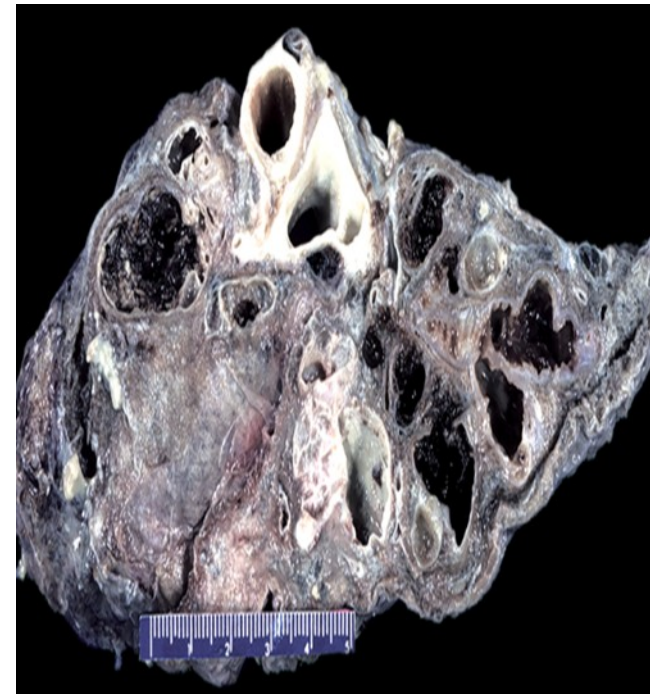


# Bronchiectasis

- ❖ It may develop in association with a variety of conditions, including
  - cystic fibrosis
  - intralobar sequestration of the lung
  - immunodeficiency states
  - primary ciliary dyskinesia
  - Kartagener syndromes
  - Infections, including necrotizing pneumonia caused by bacteria, viruses, or fungi;
  - Bronchial obstruction, due to tumor, foreign bodies, or mucus impaction
  - rheumatoid arthritis, SLE, IBD, COPD,
- ❖ One fourth to one half of cases are **idiopathic**.

# MORPHOLOGY

- ❖ Usually affects the **lower lobes bilaterally**
- ❖ Is most severe in the more **distal bronchi and bronchioles**.
- ❖ The airways are dilated, sometimes up to four times normal size.
- ❖ On the cut surface of the lung, the dilated bronchi appear cystic and are filled with mucopurulent secretions
- ❖ The histologic findings vary with the activity and chronicity of the disease.



# Clinical Course

- ❖ Bronchiectasis causes **severe, persistent cough; expectoration of foul smelling sputum**
- ❖ Obstructive respiratory insufficiency can lead to **marked dyspnea and cyanosis.**
- ❖ cor pulmonale, brain abscesses, and amyloidosis are less frequent **complications** of bronchiectasis **currently** than in the **past.**

# **RESTRICTIVE LUNG DISEASES**

# Chronic Diffuse Interstitial (Restrictive) Diseases

- ❖ Restrictive lung disorders occur in **two** general conditions:
  - chronic interstitial and infiltrative diseases, such as **pneumoconioses** and **interstitial fibrosis** of unknown etiology
  - chest wall disorders (e.g., neuromuscular diseases such as poliomyelitis, severe obesity, pleural diseases, and kyphoscoliosis)
- ❖ Chronic interstitial pulmonary diseases are a heterogeneous group of disorders
  - characterized predominantly by **inflammation and fibrosis** of the **pulmonary interstitium**.
- ❖ Many of the entities are of **unknown** cause
- ❖ Diffuse restrictive diseases are categorized based on **histology** and **clinical** features

**Table 15-5 Major Categories of Chronic Interstitial Lung Disease**

|                                                                                                                                                                                                                                             |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Fibrosing</b>                                                                                                                                                                                                                            |
| Usual interstitial pneumonia (idiopathic pulmonary fibrosis)<br>Nonspecific interstitial pneumonia<br>Cryptogenic organizing pneumonia<br>Connective tissue disease-associated<br>Pneumoconiosis<br>Drug reactions<br>Radiation pneumonitis |
| <b>Granulomatous</b>                                                                                                                                                                                                                        |
| Sarcoidosis<br>Hypersensitivity pneumonitis                                                                                                                                                                                                 |
| <b>Eosinophilic</b>                                                                                                                                                                                                                         |
| <b>Smoking Related</b>                                                                                                                                                                                                                      |
| Desquamative interstitial pneumonia<br>Respiratory bronchiolitis-associated interstitial lung disease                                                                                                                                       |
| <b>Other</b>                                                                                                                                                                                                                                |
| Langerhans cell histiocytosis<br>Pulmonary alveolar proteinosis<br>Lymphoid interstitial pneumonia                                                                                                                                          |

# Fibrosing Diseases

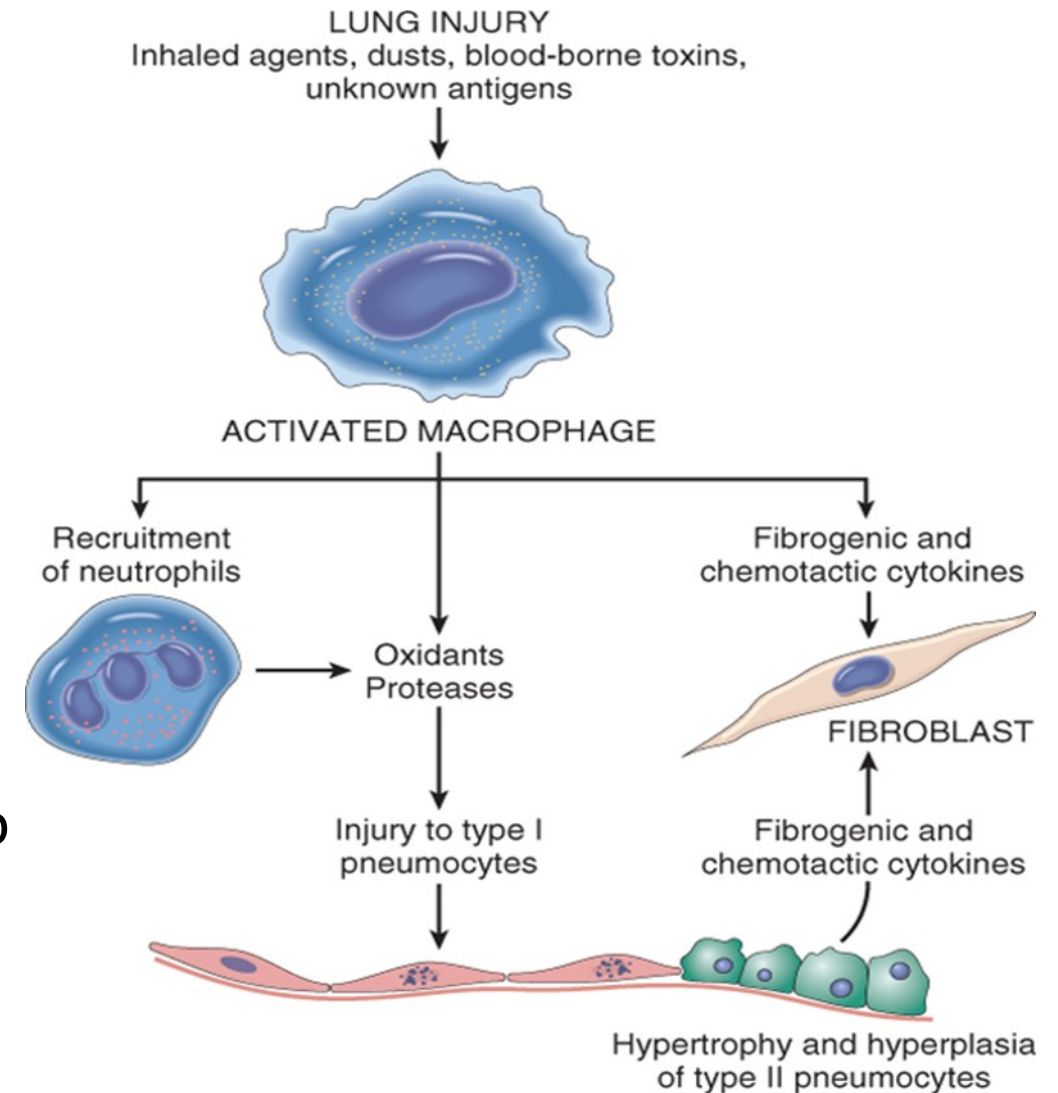
## Idiopathic Pulmonary Fibrosis

- ❖ IPF refers to a clinicopathologic syndrome marked by **progressive interstitial pulmonary fibrosis and respiratory failure**.
- ❖ In Europe the term *cryptogenic fibrosing alveolitis* is more popular.
- ❖ IPF has characteristic radiologic, pathologic, and clinical features.



# Pathogenesis

- ❖ Arises in **genetically predisposed** individuals
- ❖ The **implicated factors** are :
  - *Environmental factors*
    - Most important among these is **cigarette smoking**
  - *Genetic factors*
  - *Age*
    - IPF is a disease of **older individuals**
- **Pathogenesis** of IPF is **complex** and **poorly understood**.
- For example, it is **unknown** precisely how **alveolar epithelial cell damage** translates into **interstitial fibrosis**.



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# MORPHOLOGY

- ❖ Grossly, the pleural surfaces of the lung are **cobblestoned** as a result of the **retraction of scars** along the **interlobular septa**.
- ❖ Microscopically, the hallmark is **patchy interstitial fibrosis (honeycomb fibrosis)**.
- ❖ There is **mild to moderate inflammation** within the fibrotic areas,
- ❖ Foci of **squamous metaplasia** and **smooth muscle hyperplasia** may be present, along with pulmonary arterial hypertensive changes (**intimal fibrosis** and **medial thickening**).
- ❖ In acute exacerbations **diffuse alveolar damage** may be **superimposed** on these chronic changes.

❖ **Pneumoconioses** (Occupational ILD): Due to inhalation of inorganic dusts.

- ❑ **Silicosis:** Nodular fibrosis from silica dust.

- ❑ **Asbestosis:** Diffuse interstitial fibrosis from asbestos fibers; associated with pleural plaques and lung cancer.

- ❑ Coal Worker's Pneumoconiosis (Black Lung Disease).

❖ In general, only a small percentage of exposed people develop occupational respiratory diseases.

**Table 15-6** Lung Diseases Caused by Air Pollutants

| Agent                                                         | Disease                                                                                                           | Exposure                                                                   |
|---------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------|
| <b>Mineral Dusts</b>                                          |                                                                                                                   |                                                                            |
| Coal dust                                                     | Anthracosis<br>Macules<br>Progressive massive fibrosis<br>Caplan syndrome                                         | Coal mining (particularly hard coal)                                       |
| Silica                                                        | Silicosis<br>Caplan syndrome                                                                                      | Metal casting work, sandblasting, hard rock mining, stone cutting, others  |
| Asbestos                                                      | Asbestosis<br>Pleural plaques<br>Caplan syndrome<br>Mesothelioma<br>Carcinoma of the lung, larynx, stomach, colon | Mining, milling, manufacturing, and installation and removal of insulation |
| Beryllium                                                     | Acute berylliosis<br>Beryllium granulomatosis<br>Lung carcinoma (?)                                               | Mining, manufacturing                                                      |
| Iron oxide                                                    | Siderosis                                                                                                         | Welding                                                                    |
| Barium sulfate                                                | Baritosis                                                                                                         | Mining                                                                     |
| Tin oxide                                                     | Stannosis                                                                                                         | Mining                                                                     |
| <b>Organic Dusts That Induce Hypersensitivity Pneumonitis</b> |                                                                                                                   |                                                                            |
| Moldy hay                                                     | Farmer's lung                                                                                                     | Farming                                                                    |
| Bagasse                                                       | Bagassosis                                                                                                        | Manufacturing wallboard, paper                                             |
| Bird droppings                                                | Bird-breeder's lung                                                                                               | Bird handling                                                              |
| <b>Organic Dusts That Induce Asthma</b>                       |                                                                                                                   |                                                                            |
| Cotton, flax, hemp                                            | Byssinosis                                                                                                        | Textile manufacturing                                                      |
| Red cedar dust                                                | Asthma                                                                                                            | Lumbering, carpentry                                                       |
| <b>Chemical Fumes and Vapors</b>                              |                                                                                                                   |                                                                            |
| Nitrous oxide, sulfur dioxide, ammonia, benzene, insecticides | Bronchitis, asthma<br>Pulmonary edema<br>ARDS<br>Mucosal injury<br>Fulminant poisoning                            | Occupational and accidental exposure                                       |

ARDS, Acute respiratory distress syndrome.

# Granulomatous Diseases

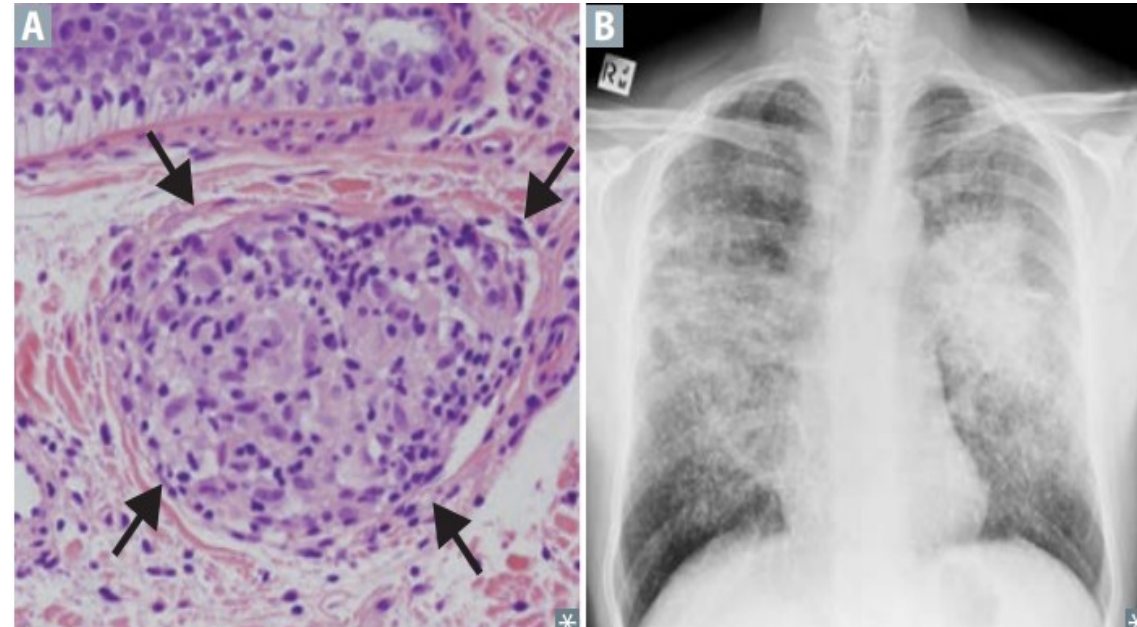
## Sarcoidosis

- A multisystem **noncaseating granulomatous** disease of **unknown** etiology
- **Bilateral hilar lymphadenopathy** or lung involvement is most common(90% )
  - **visible** on chest radiographs
- Eye and skin involvement each occurs in about 25% of cases
  - may occasionally be the presenting feature of the disease.
- The diagnosis is one of exclusion

## Sarcoidosis

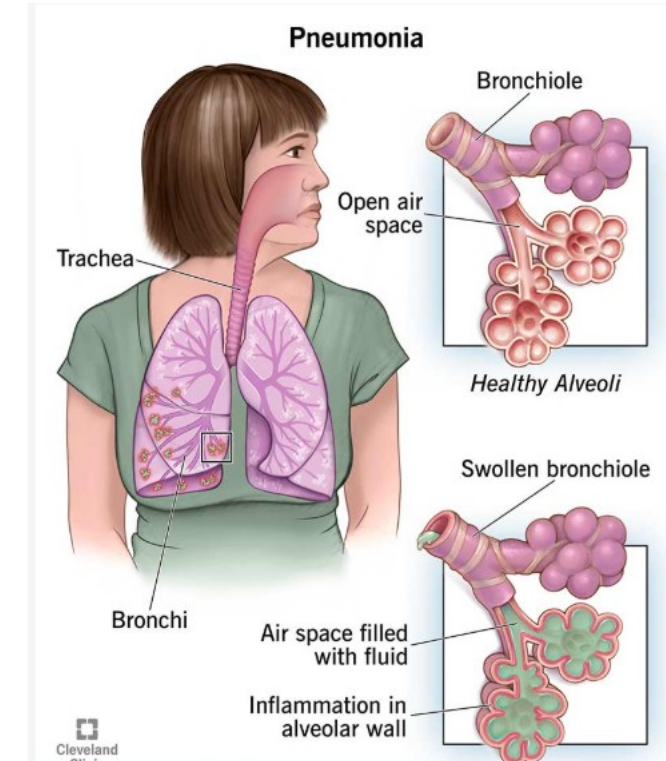
**Morphology:** regardless of the organ involved

- **Non caseating epithelioid granuloma**
- **Asteroid bodies**(inclusions in multinucleated giant cells)
- **Schaumann bodies: laminated concretions** composed of **calcium** and **proteins**
  - **not pathognomonic** of sarcoidosis
- The **lungs** are **common sites** of involvement
- **Lymph nodes** are involved in **almost all cases**
  - particularly the hilar and mediastinal nodes, but any LN



# Pneumonia

- ❖ Inflammation of the lung parenchyma
- ❖ Pneumonia can result when
  - pulmonary **defense mechanisms** are impaired
  - when the **resistance of the host** in general is impaired
- e.g.*** Immunologic deficiency, immunosuppressive agents & leukopenia
- ❖ Pulmonary defense mechanisms (clearing mechanism) can be impaired
  - Loss or suppression of **cough reflex**
    - Coma, anesthesia, neuromuscular disorders, drugs, chest pain
  - Injury to **mucociliary apparatus**
    - Cigarette smoking, viral diseases, genetic disturbances
  - Interference of action of **alveolar macrophages**
    - Alcohol, tobacco, anoxia
  - Pulmonary **congestion & edema**
  - Accumulation of **secretions**
    - Bronchiectasis, cystic fibrosis





# Classifications

- Pneumonias are classified on the basis of specific **etiologic agent** or **clinical setting**

Table 15-7 Pneumonia Syndromes

| Community-Acquired Acute Pneumonia                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | Chronic Pneumonia                                                                                                                                                                                                  |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <i>Streptococcus pneumoniae</i><br><i>Haemophilus influenzae</i><br><i>Moraxella catarrhalis</i><br><i>Staphylococcus aureus</i><br><i>Legionella pneumophila</i><br><i>Enterobacteriaceae</i> ( <i>Klebsiella pneumoniae</i> ) and <i>Pseudomonas</i> spp.<br><i>Mycoplasma pneumoniae</i><br><i>Chlamydia</i> spp. ( <i>C. pneumoniae</i> , <i>C. psittaci</i> , <i>C. trachomatis</i> )<br><i>Coxiella burnetii</i> (Q fever)<br>Viruses: respiratory syncytial virus, parainfluenza virus and human metapneumovirus (children); influenza A and B (adults); adenovirus (military recruits); | <i>Nocardia</i><br><i>Actinomyces</i><br>Granulomatous: <i>Mycobacterium tuberculosis</i> and atypical mycobacteria, <i>Histoplasma capsulatum</i> , <i>Coccidioides immitis</i> , <i>Blastomyces dermatitidis</i> |
| Health Care-Associated Pneumonia                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | Necrotizing Pneumonia and Lung Abscess                                                                                                                                                                             |
| <i>Staphylococcus aureus</i> , methicillin-sensitive<br><i>Staphylococcus aureus</i> , methicillin-resistant<br><i>Pseudomonas aeruginosa</i><br><i>Streptococcus pneumoniae</i>                                                                                                                                                                                                                                                                                                                                                                                                                | Anaerobic bacteria (extremely common), with or without mixed aerobic infection<br><i>Staphylococcus aureus</i> , <i>Klebsiella pneumoniae</i> , <i>Streptococcus pyogenes</i> , and type 3 pneumococcus (uncommon) |
| Hospital-Acquired Pneumonia                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Pneumonia in the Immunocompromised Host                                                                                                                                                                            |
| Gram-negative rods, <i>Enterobacteriaceae</i> ( <i>Klebsiella</i> spp., <i>Serratia marcescens</i> , <i>Escherichia coli</i> ) and <i>Pseudomonas</i> spp.<br><i>Staphylococcus aureus</i> (usually methicillin-resistant)                                                                                                                                                                                                                                                                                                                                                                      | Cytomegalovirus<br><i>Pneumocystis jiroveci</i><br><i>Mycobacterium avium-intracellulare</i><br>Invasive aspergillosis<br>Invasive candidiasis<br>"Usual" bacterial, viral, and fungal organisms (listed herein)   |
| Aspiration Pneumonia                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                                                                                                                                                                                                                    |
| Anaerobic oral flora ( <i>Bacteroides</i> , <i>Prevotella</i> , <i>Fusobacterium</i> , <i>Peptostreptococcus</i> ), admixed with aerobic bacteria ( <i>Streptococcus pneumoniae</i> , <i>Staphylococcus aureus</i> , <i>Haemophilus influenzae</i> , <i>Pseudomonas aeruginosa</i> )                                                                                                                                                                                                                                                                                                            |                                                                                                                                                                                                                    |

# Community acquired acute pneumonia

- ❖ Acquired from the **normal environment**
  - bacterial or viral
- ❖ Bacterial infections often follows an **URTI viral** infection
- ❖ Bacterial invasion of the lung parenchyma causes
  - **alveoli** to be filled with an inflammatory **exudate**, causing **consolidation (solidification)** of the pulmonary tissue

# Community acquired acute pneumonia

- ❖ Predisposing conditions include
  - extremes of **age**
  - **Chronic diseases** (congestive heart failure, COPD, and diabetes),
  - Congenital or acquired **immune deficiencies**
  - **Decreased** or **absent** splenic function

## Causative bacteria

- **Streptococcus pneumoniae**
- **H. influenza**
- **S. aureus**
- **K. pneumoniae**
- **M. catarrhalis**
- **Pseudomonas spp**

# Morphology

- ❖ Bacterial pneumonia has **two patterns** of **anatomic distribution**:
  - **Bronchopneumonia**
  - **Lobar pneumonia**
- ❖ **Patchy** consolidation of the lung is characteristic of **bronchopneumonia**
- ❖ The **patchy** involvement may become confluent, **producing total lobar consolidation**
- ❖ **Lobar pneumonia** results in
  - fibrinosuppurative consolidation of **large portion of a lobe** or of an **entire lobe**

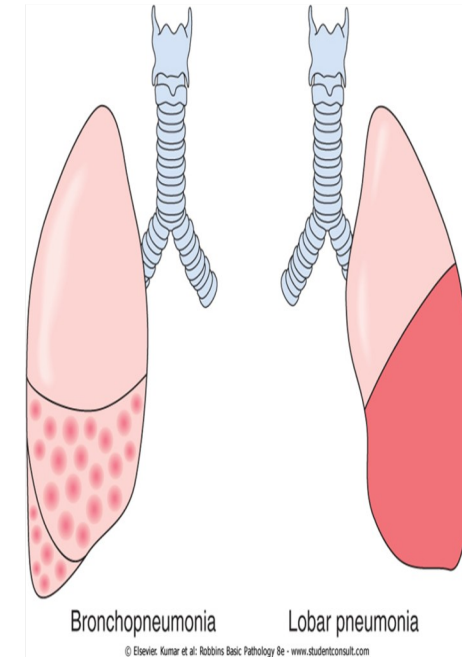


Figure 15-33 Bronchopneumonia. Section of lung showing patches of consolidation (arrows).



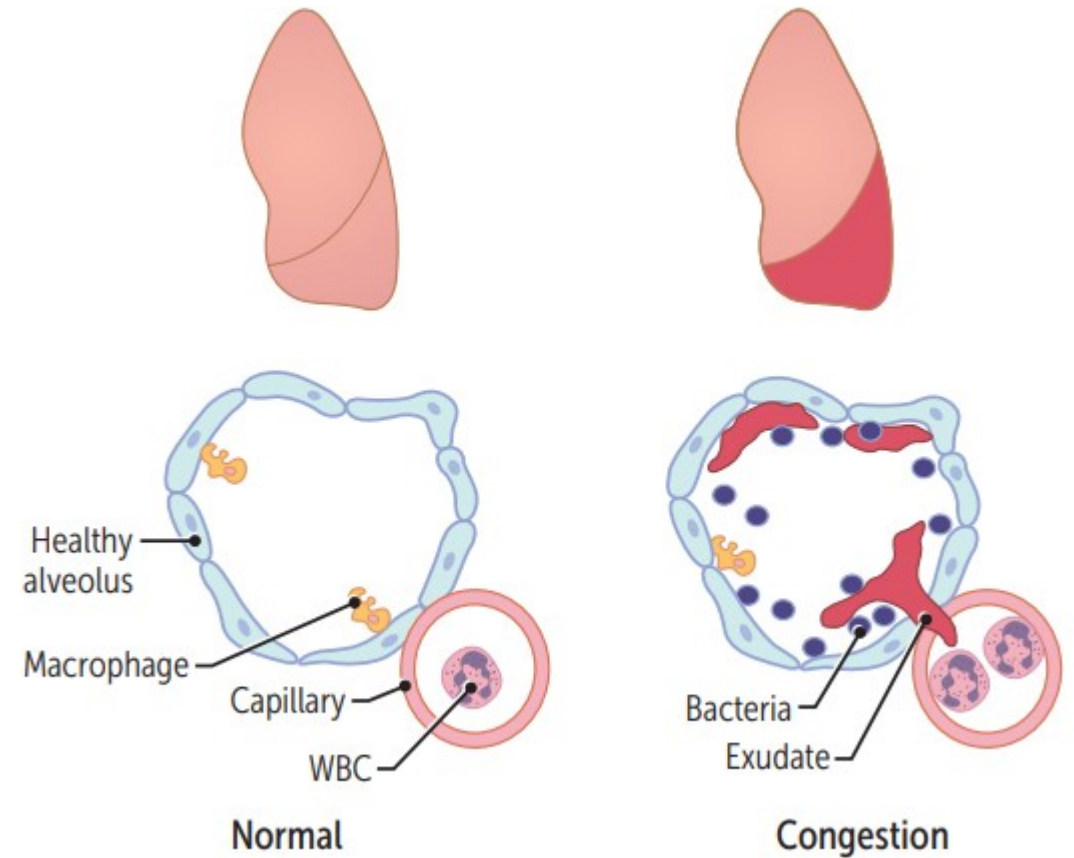
Figure 15-34 Lobar pneumonia—gray hepatization. The lower lobe is uniformly consolidated.

## Ctd...

- ❖ One organism can produce either of the two
- ❖ These **anatomic categorizations** may be difficult to apply in individual cases because **patterns overlap**
- ❖ In **lobar pneumonia**, **four stages** of the inflammatory responses have been described
  - Congestion
  - Red hepatization
  - Gray hepatization
  - Resolution

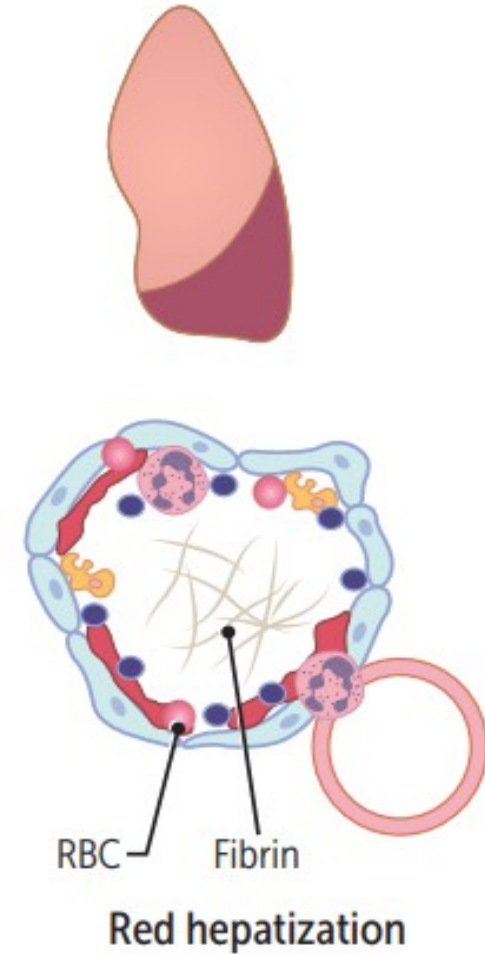
# Congestion

- ❖ Lung is heavy, red & boggy
- ❖ Vascular engorgement
- ❖ Intra alveolar fluid with **few neutrophils**
- ❖ Presence of **numerous bacteria**



# Red hepatization

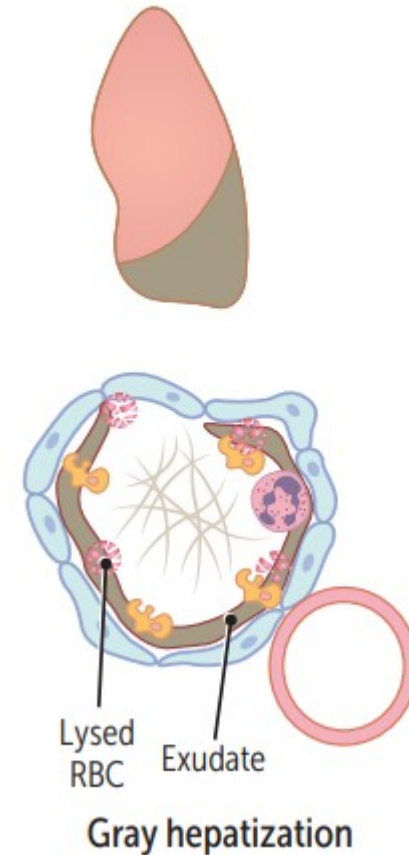
- ❖ **Massive confluent exudation** with
  - red cells
  - neutrophils
  - fibrin filling alveolar spaces
- **Grossly**, the **lobe** appears as red, firm, & airless, with a **liver like consistency**
  - Hence the term **hepatization**.





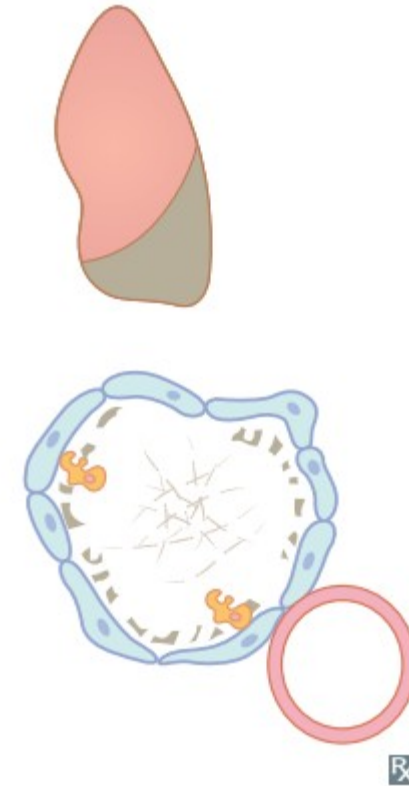
# Gray hepatization

- ❖ Marked by
  - **progressive disintegration of red cells**
  - the persistence of a fibrinosuppurative exudate
- ❖ Results in a color change to **grayish-brown**

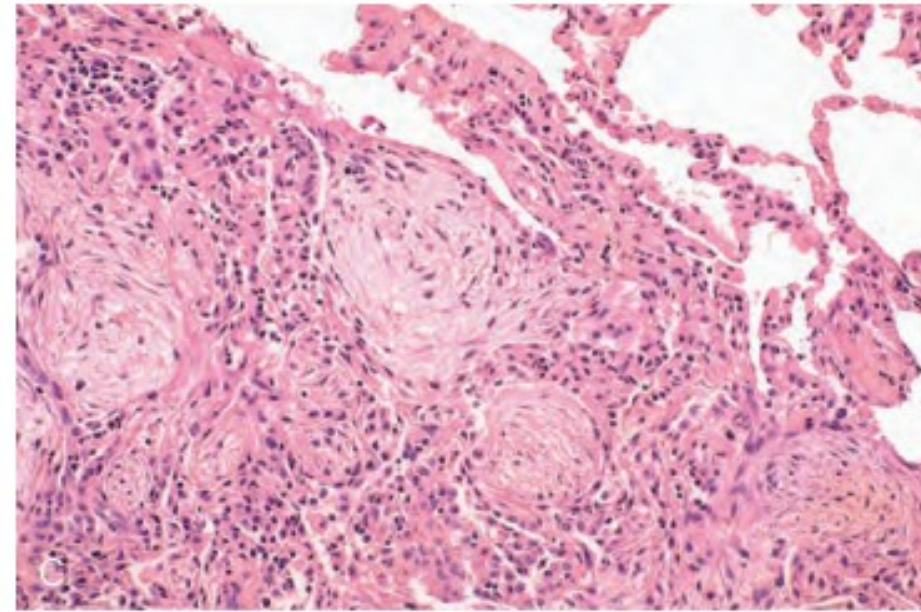
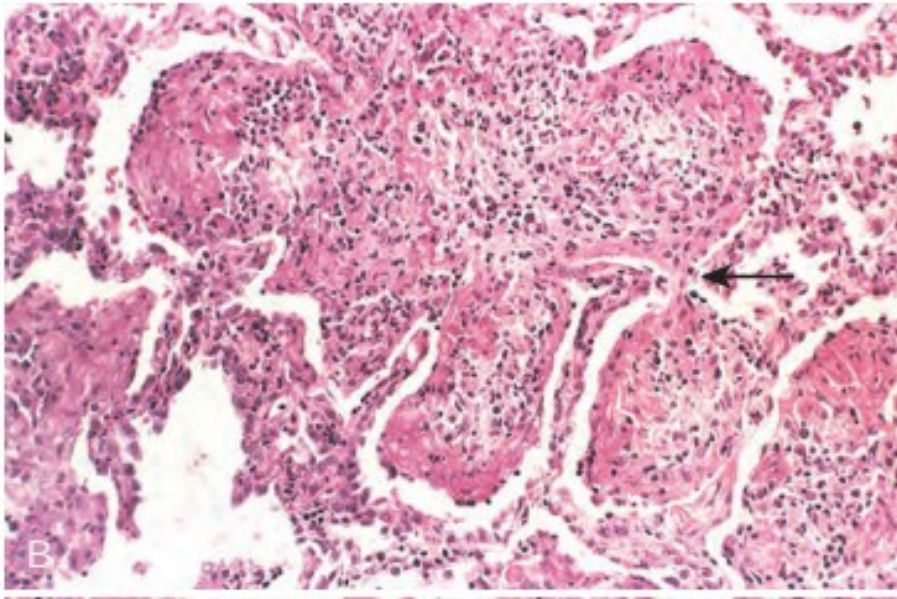
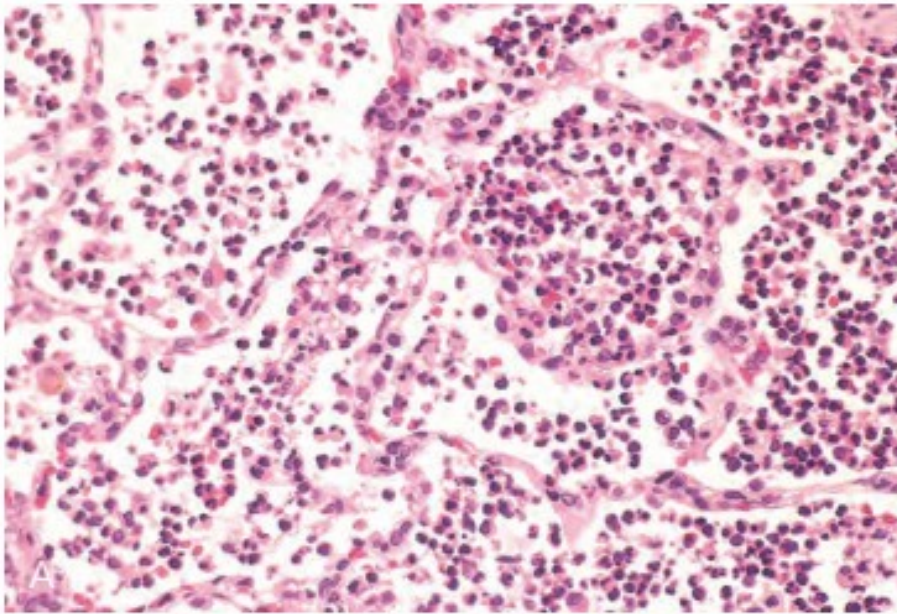


# Resolution

- ❖ Final stage => **resolution**
- ❖ The exudate **within the alveolar** spaces is broken **down by enzymatic digestion** to **produce** granular, semifluid debris that is
  - resorbed
  - ingested by macrophages
  - expectorated, or
  - organized by **fibroblasts** growing into it



Resolution



**Figure 15-35 Stages of bacterial pneumonia. A,** Acute pneumonia. The congested septal capillaries and numerous intra-alveolar neutrophils are characteristic of early red hepatization. Fibrin nets have not yet formed. **B,** Early organization of intra-alveolar exudate, seen focally to be streaming through the pores of Kohn (*arrow*). **C,** Advanced organizing pneumonia. The exudates have been converted to fibromyxoid masses rich in macrophages and fibroblasts.

# Complications

## ❖ Complications of pneumonia include

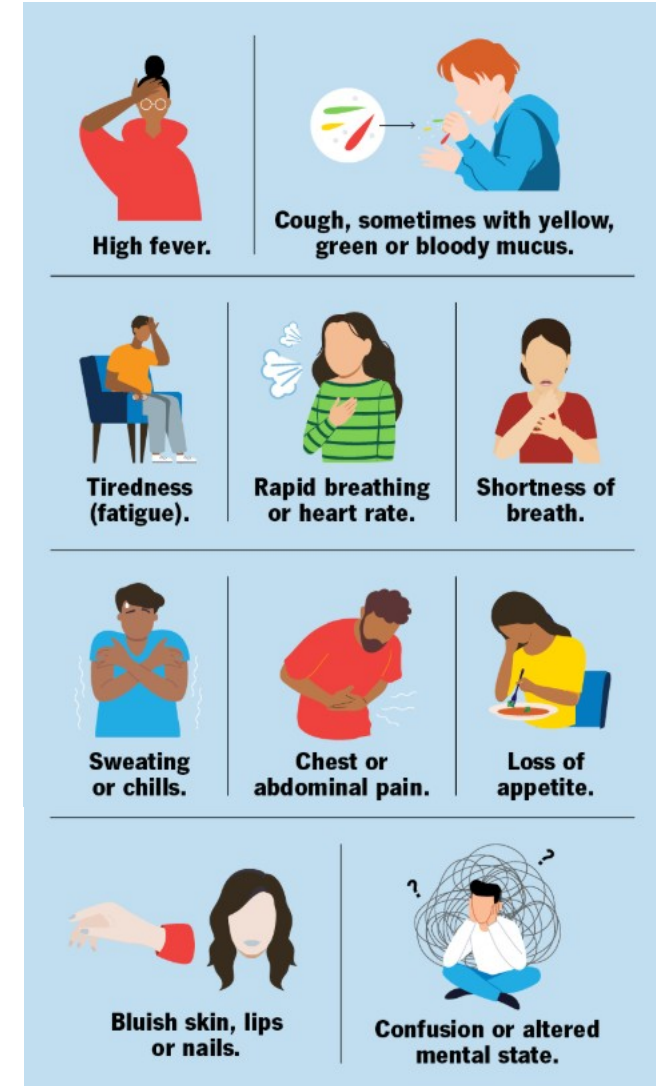
- Tissue destruction and necrosis, causing **abscess formation**
  - particularly common with **type 3 pneumococci** or *Klebsiella* infections
- **Spread** of infection to the **pleural cavity**, causing the intrapleural fibrinosuppurative reaction known as **empyema**
- **bacteremic dissemination** to
  - heart valves, pericardium
  - brain, kidneys, spleen, or joints

=> **metastatic abscesses, endocarditis, meningitis, or suppurative arthritis.**



# Clinical Course of typical pneumonia

- ❖ **Sudden onset** of high-grade fever and chills.
- ❖ **Productive cough** with purulent (yellow/green) or "rust-colored" sputum (classic for pneumococcal pneumonia).
- ❖ Pleuritic chest pain (sharp, worse on inspiration).
- ❖ Dyspnea.
- ❖ **Signs:** Tachypnea, tachycardia, dullness to percussion, **bronchial breath sounds, egophony** (E to A change), **crackles/rales**.
- ❖ Fewer than **10%** of patients with pneumonia severe enough to merit hospitalization now succumb.
- ❖ In most such instances death results from a complication, such as
  - empyema, meningitis, endocarditis, or pericarditis



# Atypical pneumonia

- ❖ Characterized by **patchy** inflammatory changes in the lungs
  - largely confined to the **alveolar septa** & pulmonary **interstitium**
    - Alveolar septa are widened & edematous, infiltrated by lymphocytes, histiocytes, plasma cells
- ❖ Atypical – denotes
  - Prominent extrapulmonary symptoms: headache, myalgias, fatigue, sore throat.
  - Dry, non-productive cough.
  - No physical findings of consolidation
  - Only moderate elevation of white cell count
  - Lack of alveolar exudate
- ❖ Causes are **mycoplasma, viruses, chlamydia**

# Nosocomial pneumonia

- ❖ Defined as pulmonary infections acquired in the **course of a hospital stay**
- ❖ Common in pts with
  - severe underlying disease
  - prolonged antibiotic therapy
  - immunosuppression
  - invasive access devices such as intravascular catheters
- ❖ Patients on mechanical ventilation are at particularly high risk.
- ❖ **S. aureus, enterobactreciae, pseudomonas** are the commonly isolated organisms
- ❖ Nosocomial infections are **serious & life threatening** infections

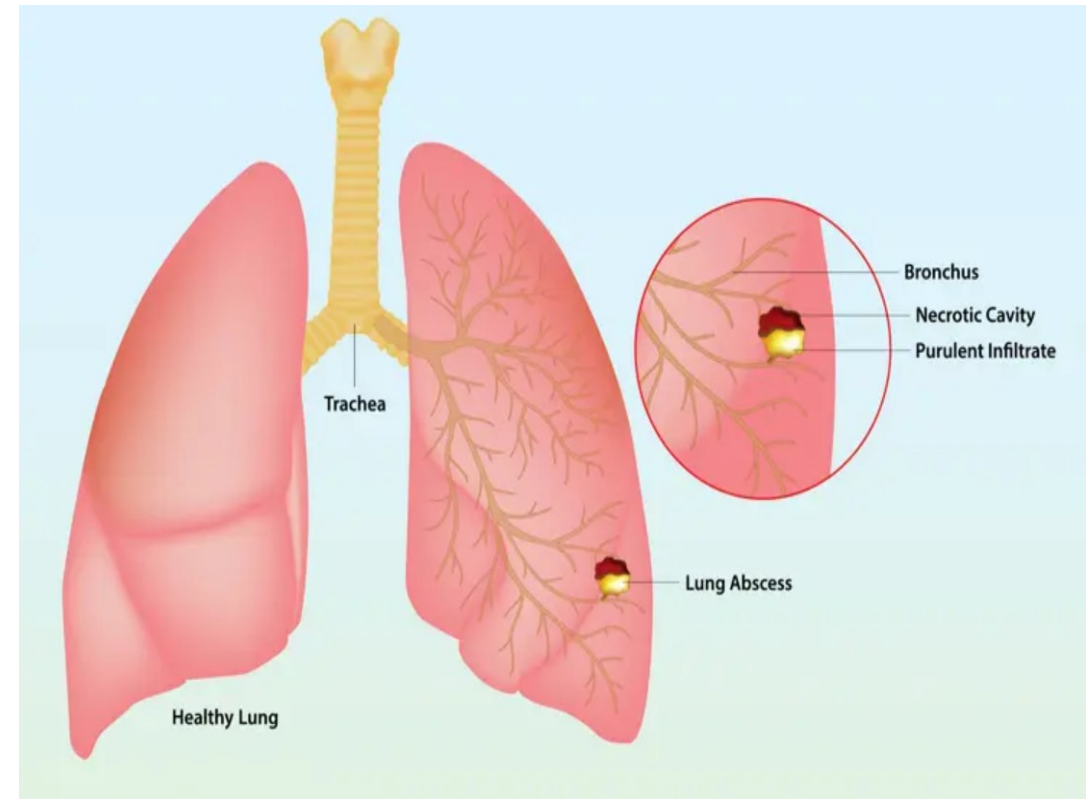


# Aspiration pneumonia

- ❖ Occurs in **markedly debilitated** pts or who **aspirate gastric contents** while
  - Unconscious (after stroke )
  - during repeated vomiting
- ❖ It is a result of **chemical inflammation & infection with normal oral flora**
- ❖ Often necrotizing, pursues a fulminant clinical course
- ❖ Frequent cause of death
- ❖ **Micro-aspiration**, in contrast, occurs frequently in almost all people
  - especially those with **GERD**
- In those who survive, **lung abscess** formation is a common complication

# Lung abscess

- ❖ A local suppurative process within the lung, characterized by
  - **necrosis** of the pulmonary tissue & formation of **cavities more than 2cm** containing necrotic debris or fluid caused by microbial infections
- ❖ This is often caused by **aspiration** which may occur during **altered consciousness**
- ❖ **Alcoholism** is the most common condition predisposing to lung abscess
- ❖ Lung abscess is considered
  - **Primary** when it resulted from **existing lung parenchymal process**
  - **Secondary** when it complicates another process  
*e.g.,* Vascular emboli or follows **rupture of extrapulmonary abscess** into the lung



# Etiologies

- ❖ Are **mixed** flora of aerobic & anaerobic
- ❖ Aerobic – S. aureus, klebsiella, haemophilus, pseudomonas, nocardia, E. coli, streptococcus, mycobacteria
- ❖ Oral cavity anaerobes – bacteroids, **fusobacterium & peptococcus spp**
- ❖ Fungi – candida, aspergillus
- ❖ Parasites – E. histolytica

# Morphology

- ❖ Abscesses vary in diameter from a few millimeters to large cavities of 5 to 6 cm
- ❖ Due to aspiration, more common on the **right** and **single**
- ❖ Abscess in the case of **bronchiectasis** & **pneumonia** are usually **multiple & basal**
- ❖ **Septic emboli** and **pyemic abscesses** are **multiple** and may affect any region of the lungs.
- ❖ The cardinal histologic change in all abscess is
  - Suppurative destruction of lung parenchyma within the central area of cavitations

# Lung abscess

## Clinical course

- ❖ Clubbing of the fingers & toes
- ❖ It is always important to exclude **cancer** whenever there is lung abscess
- ❖ Can be Complicated by
  - Septic emboli
  - infection of pleural cavity
  - brain abscess
  - rarely amyloidosis



# Chronic pneumonia

- ❖ Localized lesion in the **immunocompetent** pts
  - with or without regional lymph node involvement
- ❖ Typically, the inflammatory reaction is **granulomatous** , and is caused by
  - Bacteria (e.g., *M. tuberculosis*)
  - Fungi (e.g., *Histoplasma capsulatum*).

# Pneumonia in the immunocompromised host

- ❖ Classified according to the etiologic agent:
  - Bacteria – *p.aeruginosa*, *mycobacteria* species
  - Viruses – CMV, herpes virus
  - Fungi – *P. carinii*, *candida*, *aspergillus*

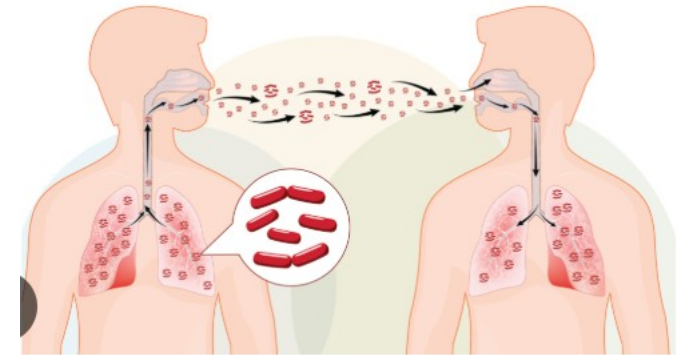


# TUBERCULOSIS

- ❖ Tuberculosis is a serious **chronic communicable granulomatous pulmonary and systemic** disease caused most often by *M. tuberculosis*.
- ❖ Usually involves the lungs but may affect **any organ or tissue** in the body.
- ❖ The source of transmission is
  - humans with **active** tuberculosis who release mycobacteria present in **sputum**
  - by drinking **milk** contaminated with *M. bovis*
- ❖ It is important that **infection** with *M. tuberculosis* be differentiated from **active** disease.
- ❖ Infection typically leads to the development of **delayed hypersensitivity** to *M. tuberculosis* antigens, which can be detected by the **tuberculin (PPD, or Mantoux) skin test**.

# Transmission

- ❖ Infections are acquired by direct person-to-person transmission of **airborne droplets**
  - generated by **expectoration**
  - by exposure to **contaminated secretions** of infected persons
- ❖ **Oropharyngeal** and **intestinal** tuberculosis contracted by drinking **milk** contaminated with ***Mycobacterium bovis***



# TUBERCULOSIS

- ❖ Infection with *M. tuberculosis* leads to the development of **delayed hypersensitivity**
  - **2 to 4 weeks** after the infection
  - Detected by the **tuberculin (Mantoux) test**.
    - ❖ Intracutaneous injection of 0.1 mL of PPD induces a visible and **palpable induration** (at least 5 mm in diameter) that peaks in **48 to 72 hours**
    - ❖ Positive tuberculin test result signifies **cell-mediated hypersensitivity** to tubercular antigens.
    - ❖ It **does not differentiate** between **infection and disease**
- ❖ False negative reactions
  - some viral infection
  - sarcoidosis, malnutrition, Hodgkin's lymphoma, immunosuppression,



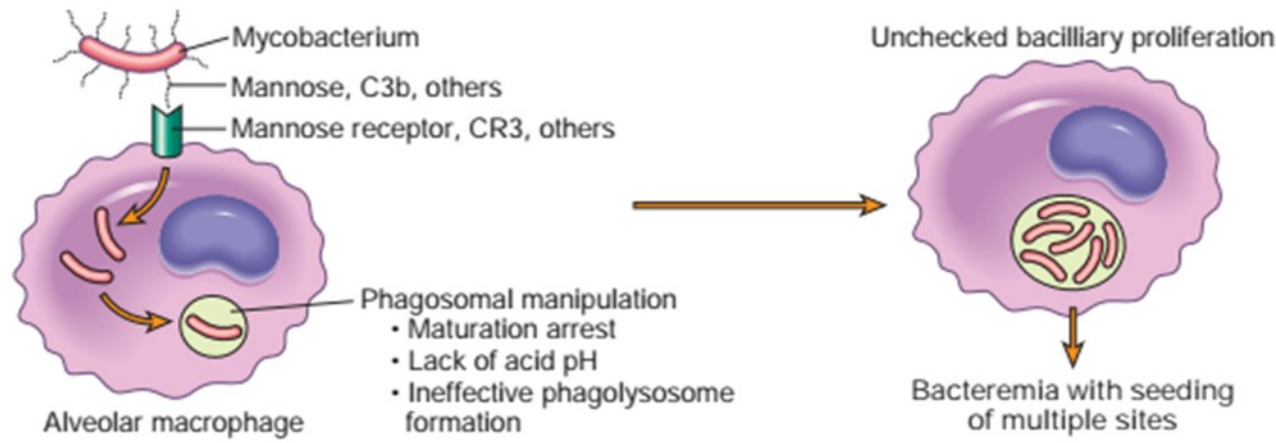
# Etiology

- ❖ Mycobacteria are **slender rods** that are **acid fast**
- ❖ *M. tuberculosis hominis* is responsible for most cases of tuberculosis
- ❖ *Mycobacterium bovis* – **oropharyngeal & intestinal** tuberculosis
- ❖ Both species are **obligate aerobes** whose slow growth is retarded by a pH lower than **6.5**.
- ❖ *M. avium-intracellulare*, are much less virulent than *M. tuberculosis*

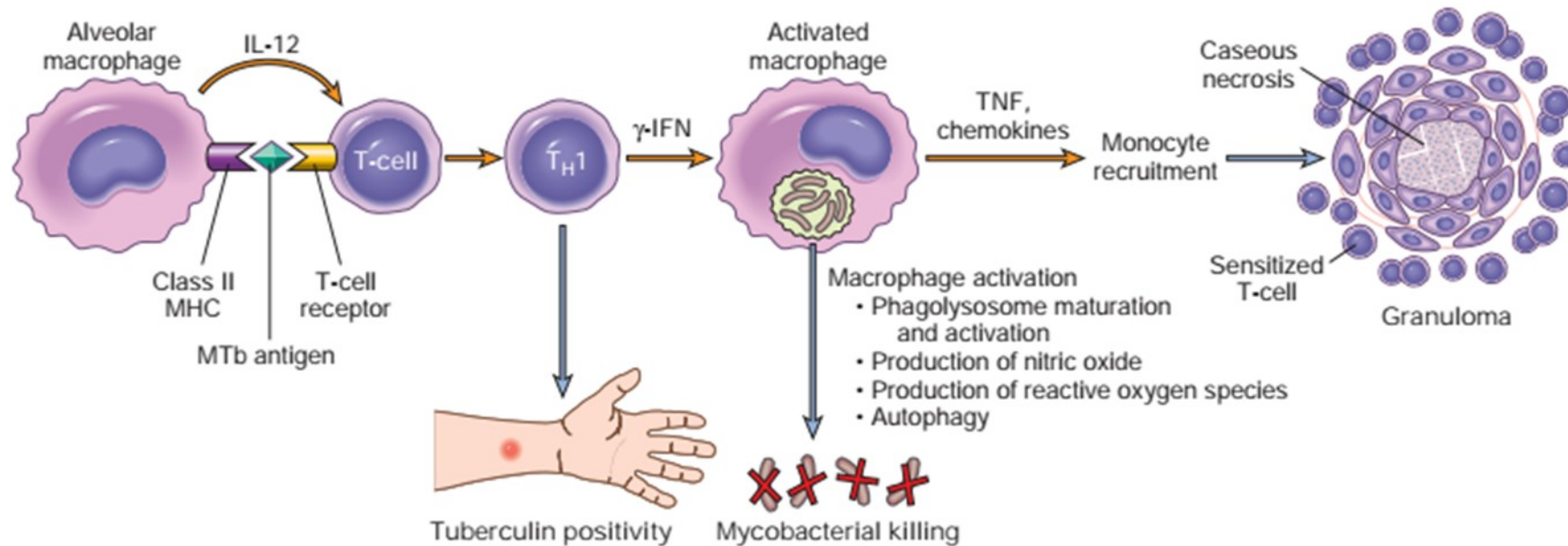
# Pathogenesis

- ❖ The outcome of infection in a previously unexposed, immunocompetent person depends on the development of **anti-mycobacterial T-cell-mediated** immunity.
- ❖ Infection by *M. tuberculosis* proceeds in steps
  - From initial infection of macrophages => to a subsequent **TH1 response** that both **contains the bacteria** and **causes tissue damage**
- ❖ Early in infection, *M. tuberculosis* replicates **essentially unchecked** within macrophages
- ❖ Later in infection, the **cell response** stimulates macrophages to contain the proliferation of the bacteria

## A. INFECTION BEFORE ACTIVATION OF CELL MEDIATED IMMUNITY



## B. INITIATION AND CONSEQUENCES OF CELL MEDIATED IMMUNITY



**Figure 8-24** The sequence of events in primary pulmonary tuberculosis, commencing with inhalation of virulent *Mycobacterium tuberculosis* organisms and culminating with the development of cell-mediated immunity to the organism. **A**, Events occurring during early infection, before activation of T-cell-mediated immunity. **B**, The initiation and consequences of T-cell-mediated immunity. The development of resistance to the organism is accompanied by the appearance of a positive tuberculin test.  $\gamma$ -IFN, interferon- $\gamma$ ; MHC, major histocompatibility complex; MTB, *M. tuberculosis*; TNF, tumor necrosis factor.

- ❖ In many people this response **halts** the infection before significant tissue destruction or illness occur.
- ❖ In other people the infection progresses due to
  - **advanced age**
  - **immunosuppression**
- ❖ The ongoing immune response results in **caseating necrosis**.
- ❖ Activated macrophages also secrete **TNF** and **chemokines**, which promote recruitment of more **monocytes**.
- ❖ Host susceptibility to disease
  - People with genetic **deficiencies** in the **IL-12 pathway** and the **IFN- $\gamma$  pathway**, are **vulnerable** to severe mycobacterial infections.
- ❖ Immunological state in active tuberculosis
  - It is not entirely clear why some people progress from latent to active tuberculosis.

## Ctd...

- ❖ The pathologic features of tuberculosis, such as
  - **Caseating granulomas** and **cavitation**, are the result of the **destructive tissue hypersensitivity** that is part of the **host immune response**.

### Clinical Features

- ❖ Clinical tuberculosis is separated into **two** important pathophysiologic types:
  - “Primary” tuberculosis, which occurs in the **non-immune host**
  - “Secondary” tuberculosis, which occurs in the host who is **immune** to *M. tuberculosis*.

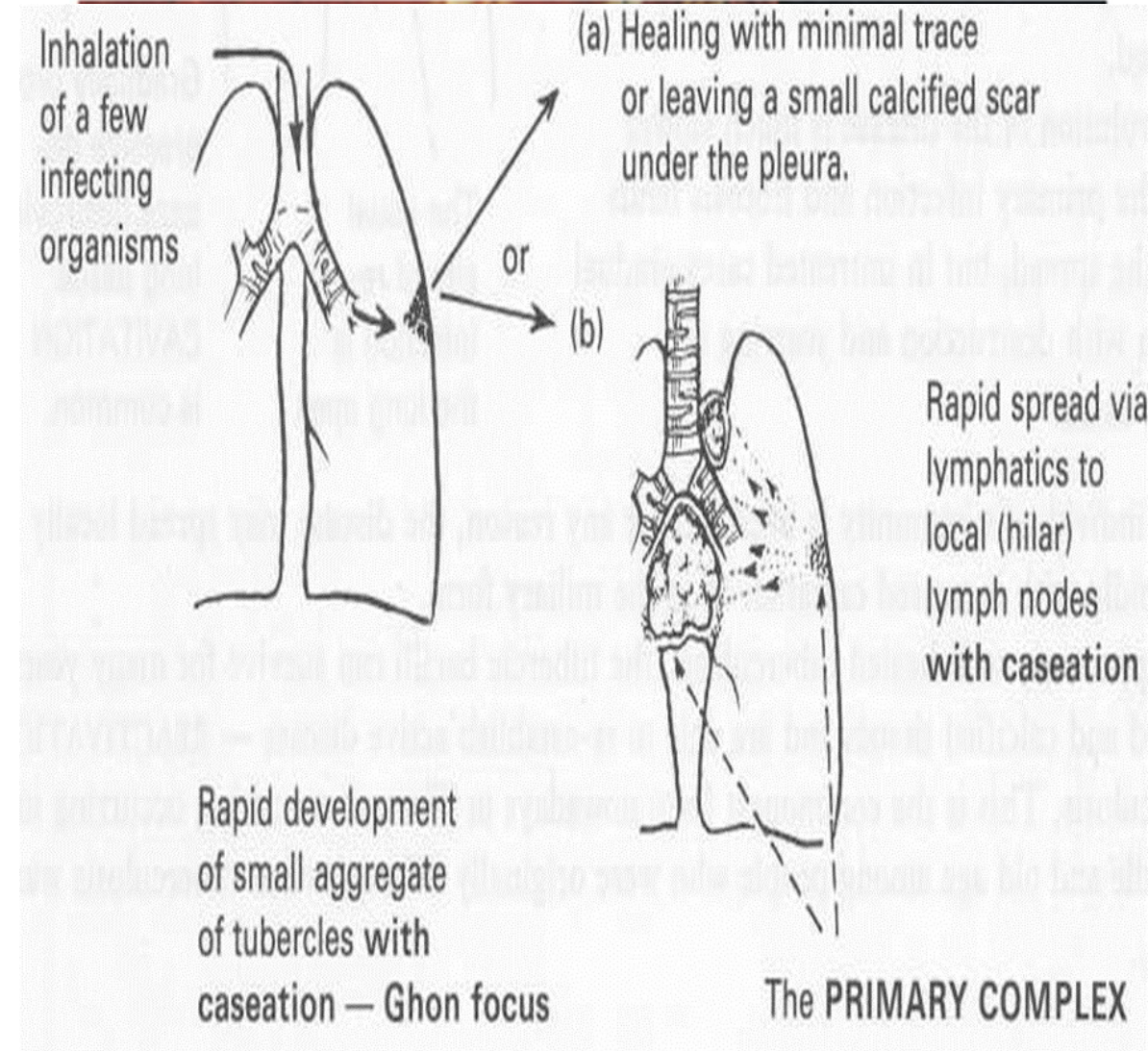


# Primary Tuberculosis

- ❖ The form of disease that develops in a **previously unexposed**, and therefore **unsensitized**, person.
- ❖ The source of the organism is **exogenous**.
- ❖ About **5%** of those **newly infected** develop significant disease.
- ❖ In most people, the primary infection is **contained**, but in others, primary tuberculosis is progressive.
- ❖ The diagnosis of progressive primary tuberculosis in adults can be difficult.
- ❖ In contrast to secondary tuberculosis (**apical disease** with **cavitation**)
  - Progressive primary tuberculosis more often resembles an ***acute bacterial pneumonia*** with consolidation of the lobe, hilar adenopathy, and pleural effusion.
- ❖ **Lymphohematogenous dissemination** following primary infection may result in the development of **tuberculous meningitis** and **miliary tuberculosis**

# Morphology

- ❖ Primary tuberculosis almost always begins in the **lungs**.
- ❖ The inhaled bacilli implant in the **distal** airspaces of the **lower part of the upper lobe** or the **upper part of the lower lobe**, usually **close to the pleura**.
- ❖ As sensitization develops, a **1- to 1.5-cm** area of gray-white inflammatory consolidation emerges => The **Ghon focus**
  - In most cases, the center of this focus undergoes **caseous necrosis**



parenchymal focus is under the pleura in the lower part of the upper lobe (red arrow). Hilar lymph nodes with caseation are seen on the left (blue arrow).

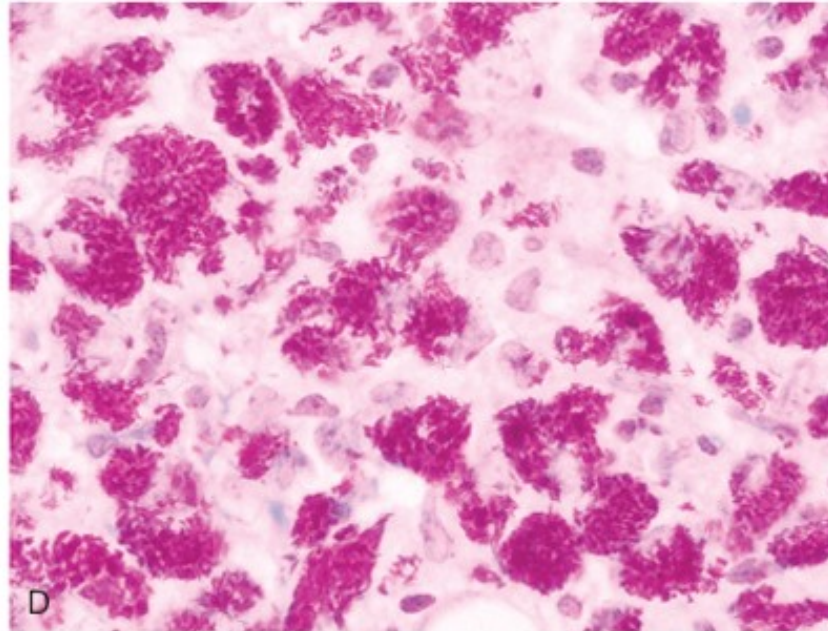
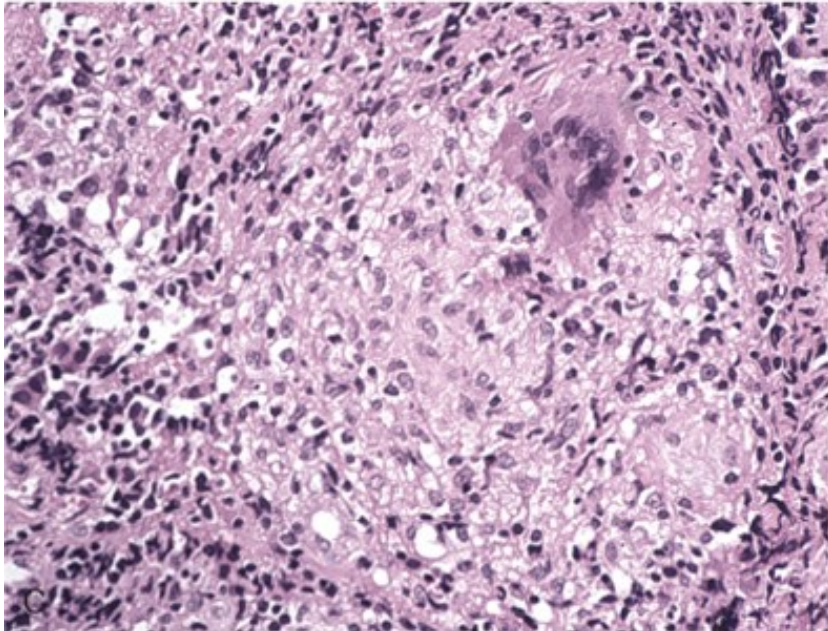
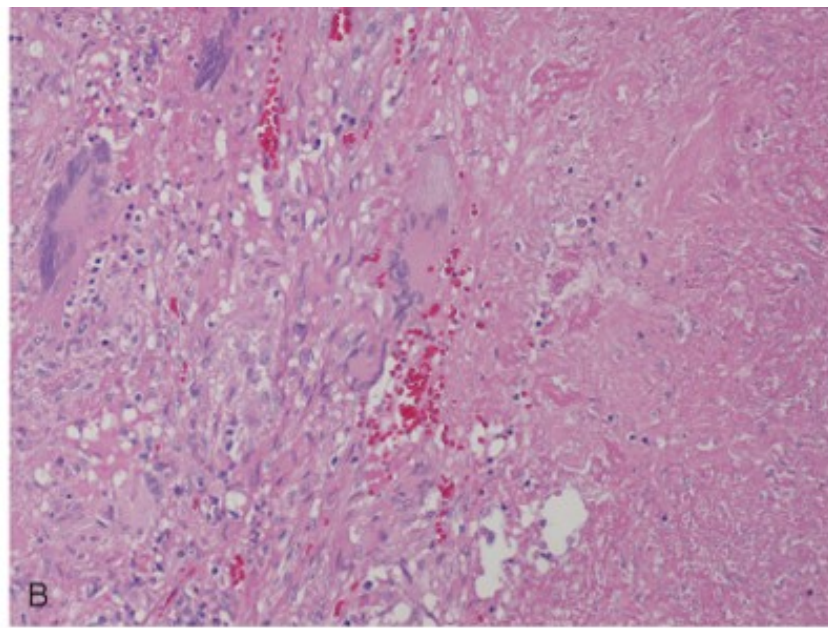
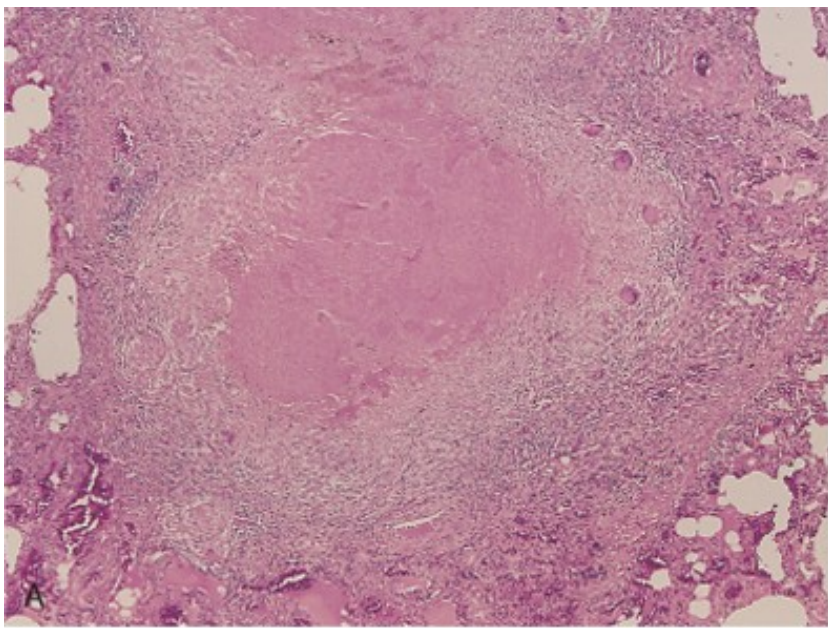
# TUBERCULOSIS

- ❖ Tubercle bacilli, either free or within phagocytes, drain to the **regional nodes**, which also often **caseate**.
- ❖ This combination of **parenchymal** lung lesion and **nodal** involvement is referred to as the **Ghon complex**.
- ❖ During the first few weeks there is also **lymphatic** and **hematogenous dissemination** to other parts of the body.
- ❖ In approximately **95%** of cases, development of **cell-mediated immunity controls** the infection.
  - Hence, the **Ghon complex** undergoes **progressive fibrosis**, often followed by radiologically detectable **calcification (Ranke complex)**.

# Histology

- ❖ Sites of active involvement are marked by
  - A characteristic **granulomatous** inflammatory reaction that forms both **caseating** and **noncaseating** tubercles .
- ❖ The granulomas are usually enclosed within a **fibroblastic rim** punctuated by **lymphocytes**.
- ❖ **Multinucleate giant cells** are present in the granulomas.
- ❖ **Immunocompromised** people **do not** form the characteristic granulomas and their macrophages contain many bacilli.







# Secondary Tuberculosis (Reactivation Tuberculosis)

- ❖ Is the pattern of disease that arises in a **previously sensitized host**.
- ❖ It may also result from **exogenous re-infection**
  - **waning** of the protection afforded by the primary disease
  - a **large inoculum** of virulent bacilli.
- ❖ Classically localized to the **apex** of one or both upper lobes.
- ❖ **Cavitation** occurs readily in the **secondary** form
- ❖ Should always be an important consideration in **HIV-positive** patients who present with pulmonary disease.

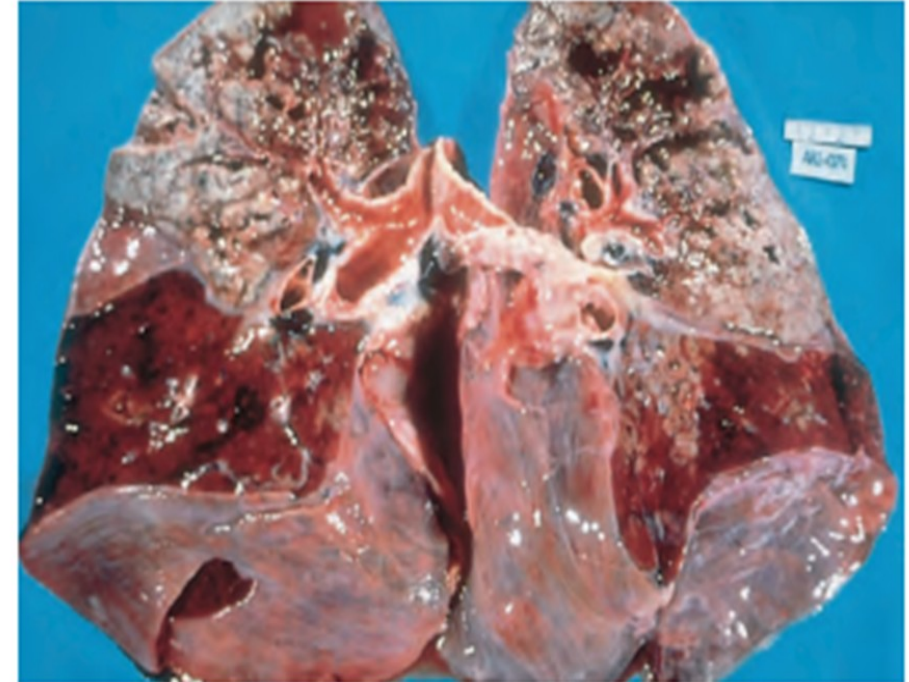
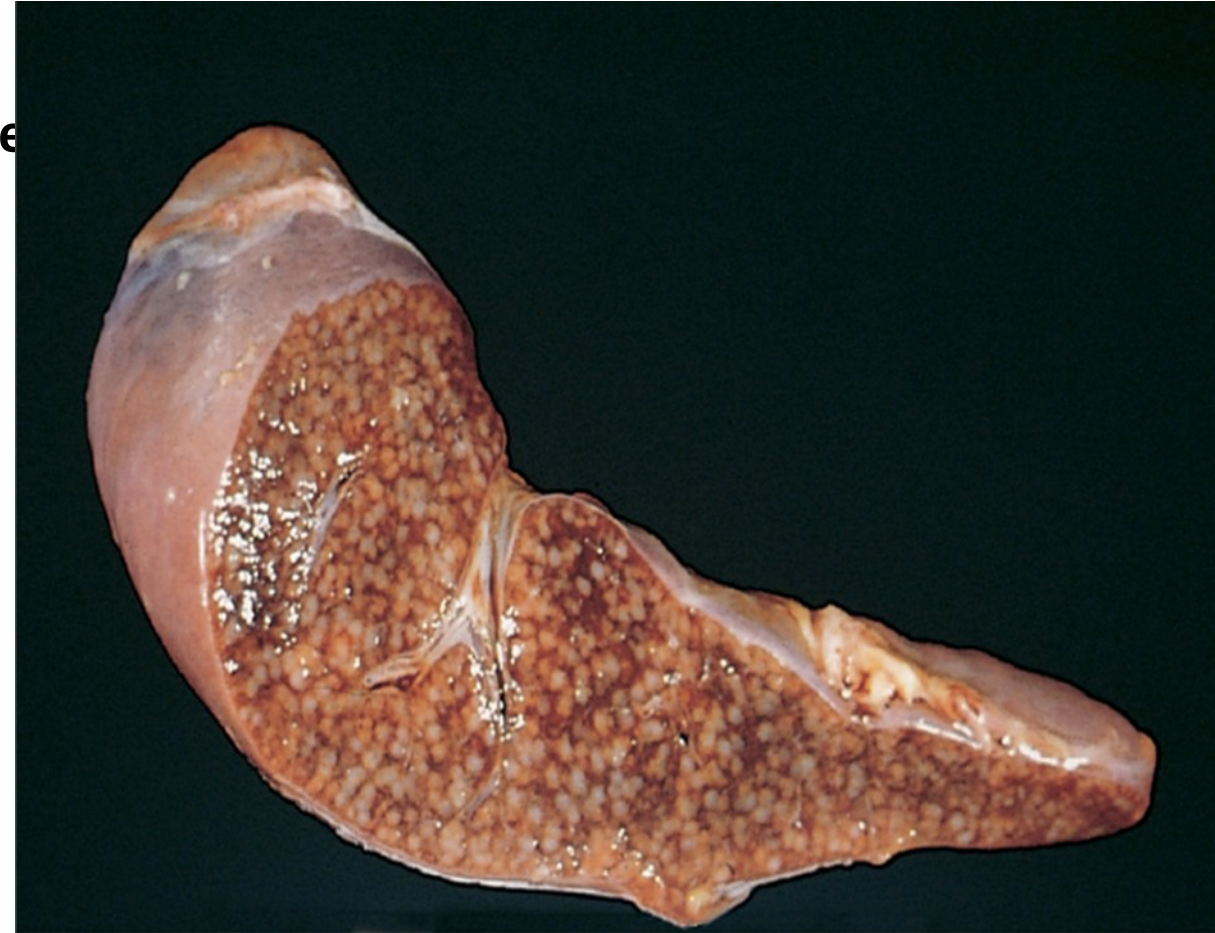


Figure 8-28 Secondary pulmonary tuberculosis. The upper parts of both lungs are riddled with gray-white areas of caseation and multiple areas of softening and cavitation.

## Miliary pulmonary disease

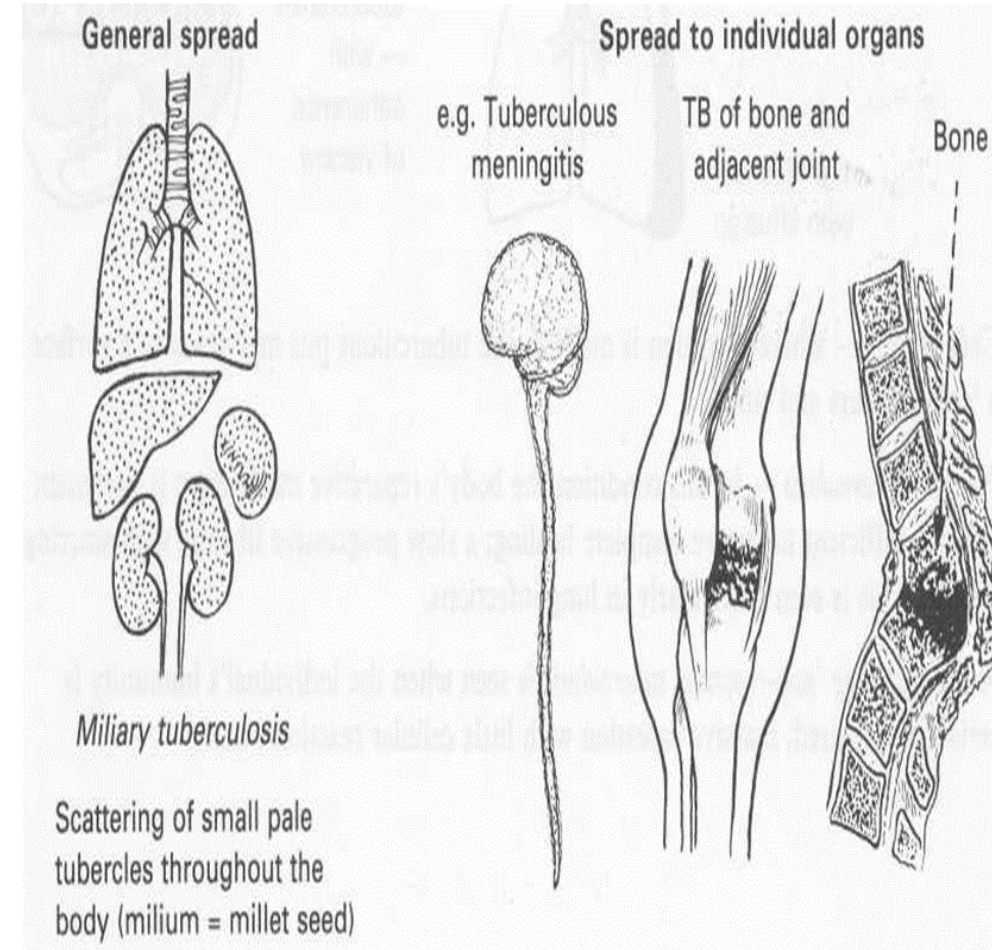
- ❖ Occurs when organisms draining through lymphatics enter the **venous blood** and **circulate back to the lung**.
- ❖ Individual lesions are either **microscopic** or **small, visible (2-mm)** foci of **yellow white** consolidation
- ❖ Miliary lesions may expand and coalesce, resulting in consolidation of **large regions** or even **whole lobes** of the lung.
- ❖ **Systemic** miliary tuberculosis occurs when bacteria disseminate through the **systemic arterial system**.
- ❖ Most prominent in the
  - **liver, bone marrow, spleen, adrenals, meninges, kidneys, fallopian tubes, and epididymis**



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# Isolated tuberculosis

- ❖ Organs that are commonly involved include
  - Meninges (tuberculous meningitis)
  - Kidneys (renal tuberculosis)
  - Adrenals
  - Bones (osteomyelitis)
  - Fallopian tubes (salpingitis)
- ❖ When the **vertebrae** are affected, the disease is referred to as **Pott disease**.
- ❖ **Paraspinal “cold” abscesses** - may track along tissue planes and present as an **abdominal** or **pelvic mass**.





# TUBERCULOSIS

- ❖ **Lymphadenitis** is the **most frequent** presentation of **extrapulmonary** tuberculosis, usually occurring in the **cervical** region (“**scrofula**”).
- ❖ In **HIV-negative** individuals, lymphadenitis tends to be **unifocal and localized**.
- ❖ HIV-positive people, on the other hand,
  - almost always have **multifocal** disease
  - systemic symptoms
  - either pulmonary or other organ involvement by active tuberculosis



# TUBERCULOSIS

## Intestinal tuberculosis

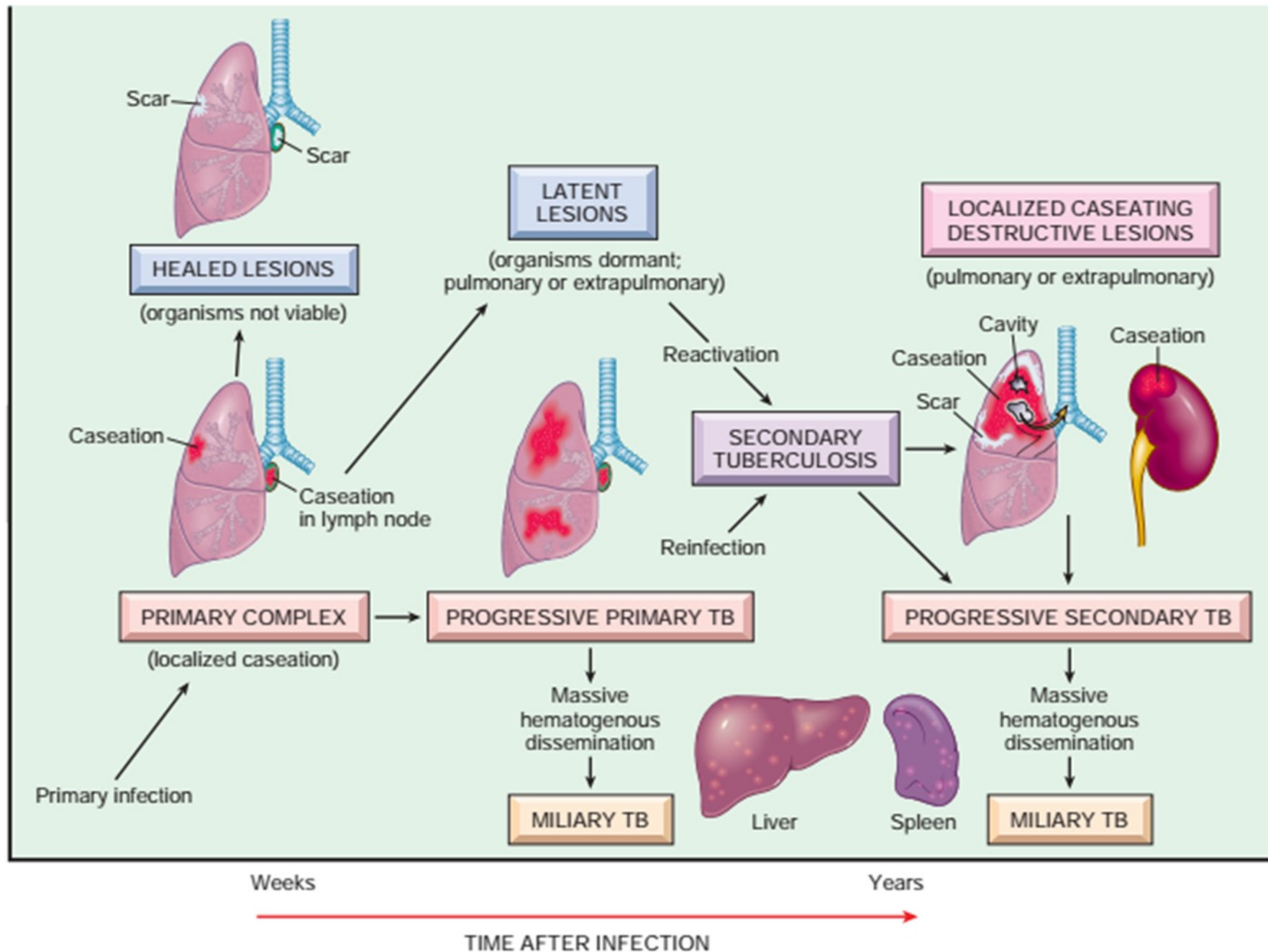
- ❖ Contracted by the drinking of **contaminated milk** is common in countries where **bovine** tuberculosis is present and **milk is not pasteurized**.
- ❖ In countries where milk is pasteurized, intestinal tuberculosis is more often caused by the **swallowing of coughed-up infective material** in patients with advanced pulmonary disease.
- ❖ Typically the organisms are seeded to mucosal lymphoid aggregates of the small and large bowel
  - which then undergo **granulomatous inflammation** that can lead to ulceration of the overlying mucosa, particularly in the **ileum**.
- Healing creates **strictures**.

# Clinical Course

- ❖ Localized secondary tuberculosis may be **asymptomatic**.
- ❖ Usually ***insidious*** in onset
- ❖ There is gradual development of both **systemic** and **localizing** symptoms.
- ❖ Systemic symptoms, probably related to **cytokines** released by activated macrophages (e.g., TNF and IL-1)
  - often appear early in the course and include **malaise, anorexia, weight loss, and fever**.

## Ctd...

- ❖ The fever is **low grade** and **remittent** (appearing late each afternoon and then subsiding)
- ❖ **Night sweats**
- ❖ Increasing amounts of **sputum**, at first mucoid and later purulent
- ❖ **Hemoptysis** is present in about **half** of all cases of pulmonary tuberculosis.
- ❖ **Pleuritic pain** may result from extension of the infection to the **pleural surfaces**.



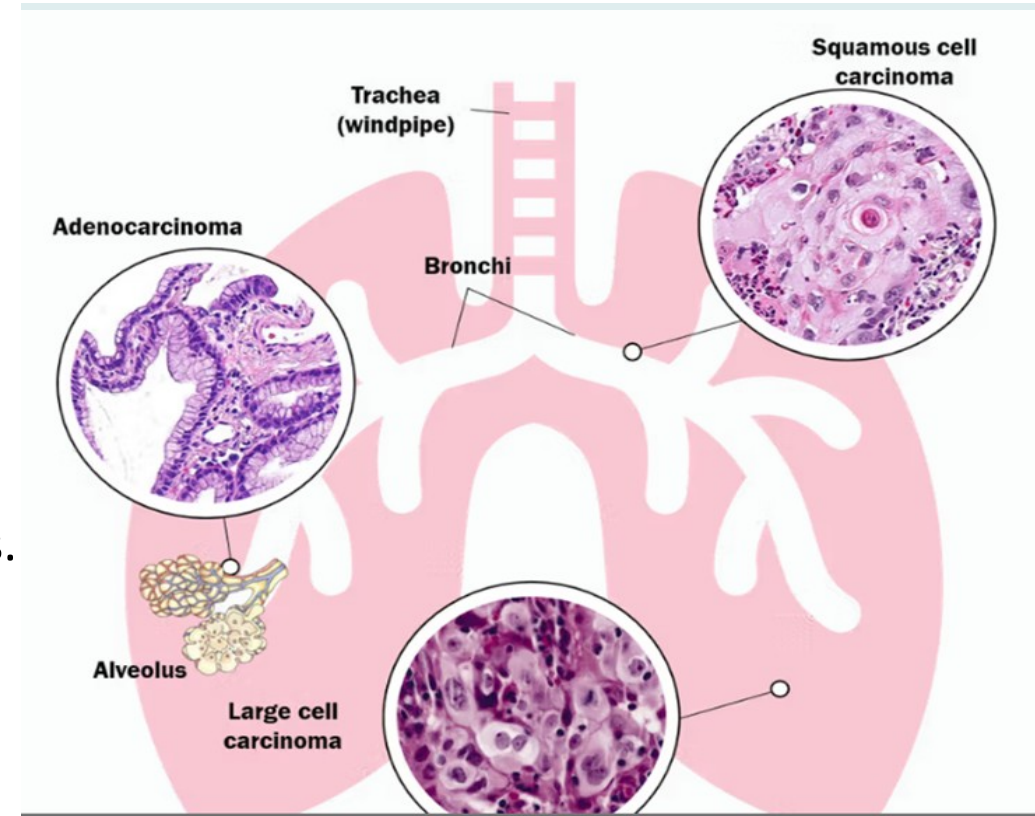
# Pulmonary Neoplasia

- ❖ Pulmonary neoplasms are broadly classified into:
  - **Primary** Lung Tumors: Arising from the lung parenchyma, bronchi, or pleura.
  - **Metastatic** Tumors: The lungs are a common site for metastases from other organs.
- ❖ Lung cancer is the **leading cause** of cancer mortality worldwide for both men and women.
- ❖ Strongest Risk Factor: **Cigarette Smoking** (responsible for **~85%** of cases). Risk correlates with **pack-years**.
  - **Other Risk Factors:**
    - Radon gas exposure
    - Asbestos exposure (synergistic with smoking; associated with both lung carcinoma and mesothelioma)
    - Occupational exposures (arsenic, nickel, chromium), Air pollution
    - Family history and genetic predisposition
    - Pulmonary fibrosis (scar carcinoma)

# Pulmonary Neoplasia

## Classification:

- ❖ **Non-Small Cell Lung Carcinoma (NSCLC) (~85%):**
  - **Adenocarcinoma:** Most common type overall; arises from glandular cells, often **peripheral**. Associated with specific mutations (**EGFR, ALK**).
  - **Squamous Cell Carcinoma: Central**, arising from the bronchial epithelium; linked to keratin pearl formation and **cavitation**.
  - **Large Cell Carcinoma:** Undifferentiated, poor prognosis.
- ❖ **Small Cell Lung Carcinoma (SCLC) (~15%):**
  - Highly malignant, **central**, and strongly associated with smoking.
  - Arises from neuroendocrine cells
  - Often presents with **paraneoplastic syndromes**.





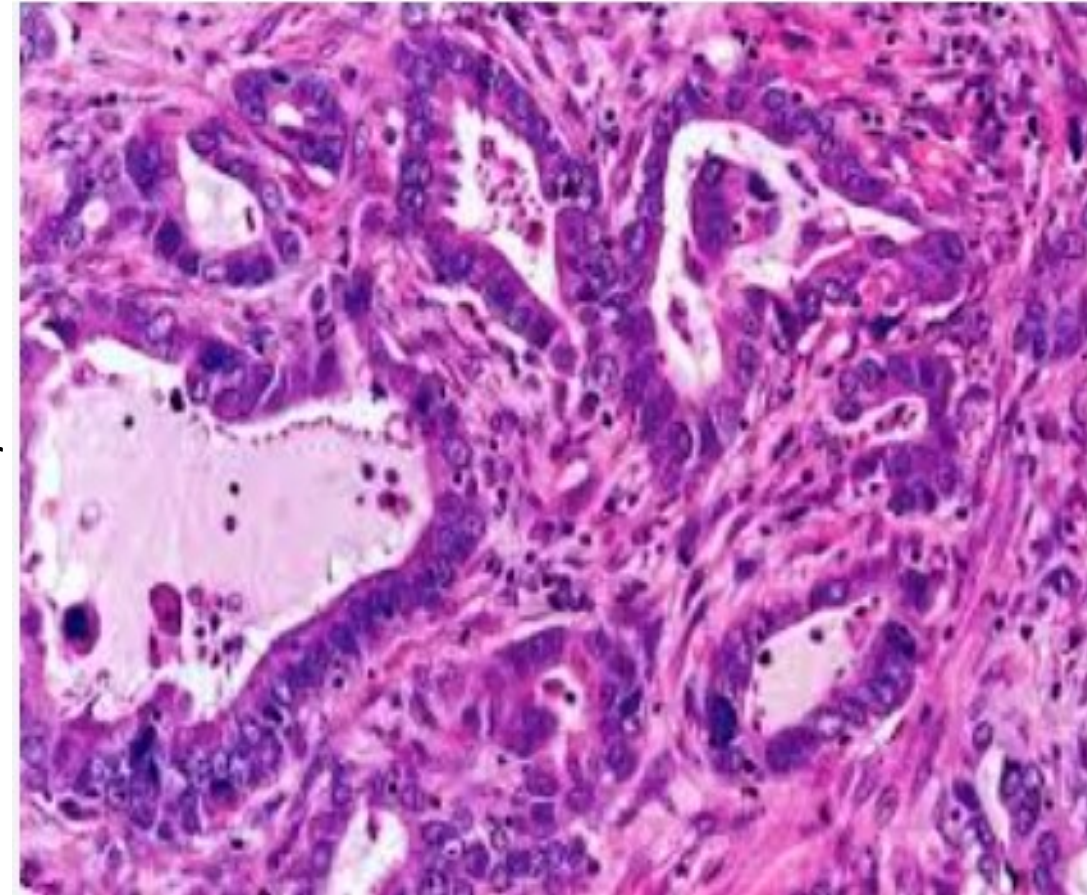
# Non-Small Cell Lung Carcinoma (NSCLC)

## A. Adenocarcinoma (Most Common type overall, ~40%)

- ❖ Origin: Bronchiolar or alveolar epithelium (Type II pneumocytes, Clara cells).
- ❖ Location: Often **peripheral**.
- ❖ Gross: Gray-white mass, often with stellate borders or pleural retraction.

### **Histopathology:**

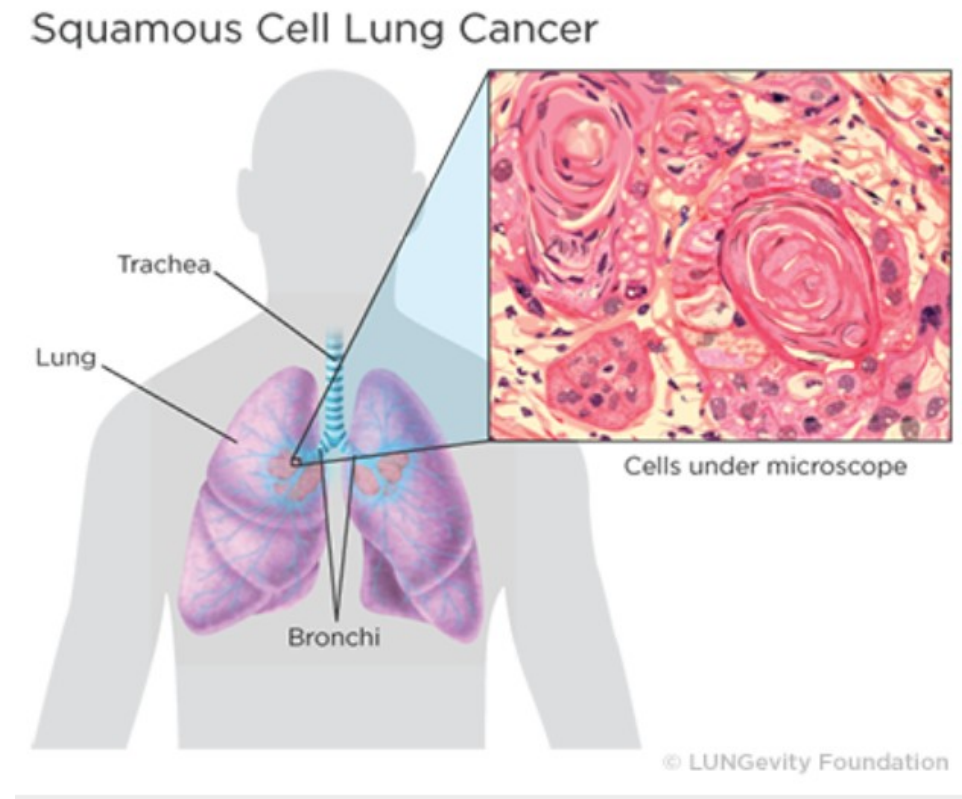
- ❖ **Lepidic Pattern:** Tumor cells grow along pre-existing alveolar structures (non-invasive). Pure lepidic growth = Adenocarcinoma in situ (AIS).
- ❖ **Acinar and Papillary Patterns:** Most common invasive patterns.
- ❖ **Micropapillary and Solid Patterns:** Associated with worse prognosis.
- ❖ **Cytology:** Tumor cells are columnar or cuboidal, with cytoplasmic mucin vacuoles.



# Non-Small Cell Lung Carcinoma (NSCLC)

## B. Squamous Cell Carcinoma (~20-30%)

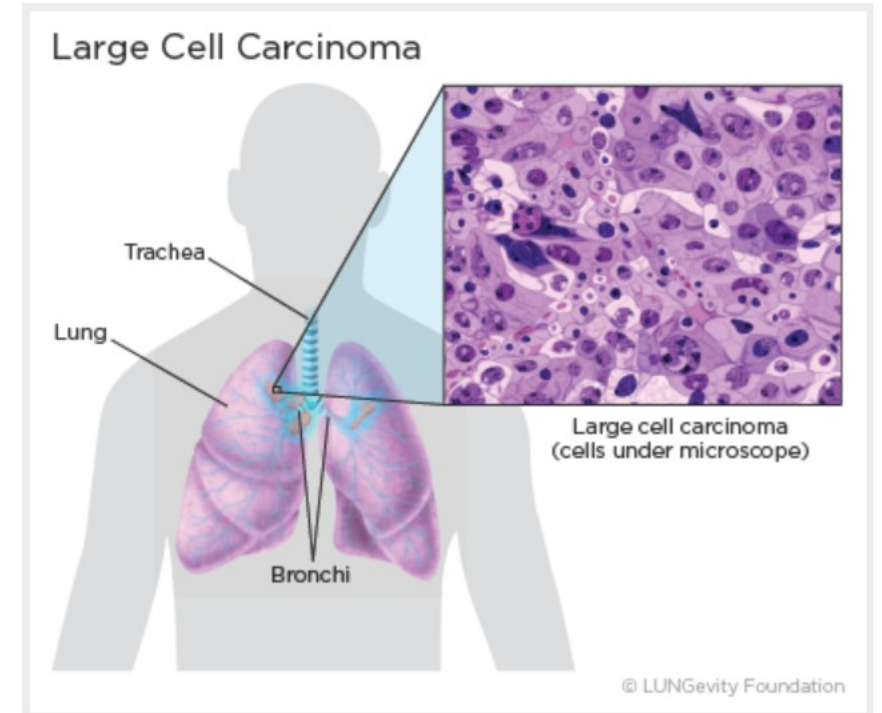
- ❖ **Origin:** Bronchial epithelium.
- ❖ **Location:** Often **central** (hilar), arising in a mainstem bronchus.
- ❖ **Gross:** Central, polypoid mass often causing airway obstruction; may **cavitate**.
- ❖ **Histopathology:**
  - Shows **keratinization** (individual cell keratinization, keratin pearls) and/or **intercellular bridges**.
  - Variants: Keratinizing, non-keratinizing, basaloid.



# Non-Small Cell Lung Carcinoma (NSCLC)

## D. Large Cell Carcinoma (~5-10%)

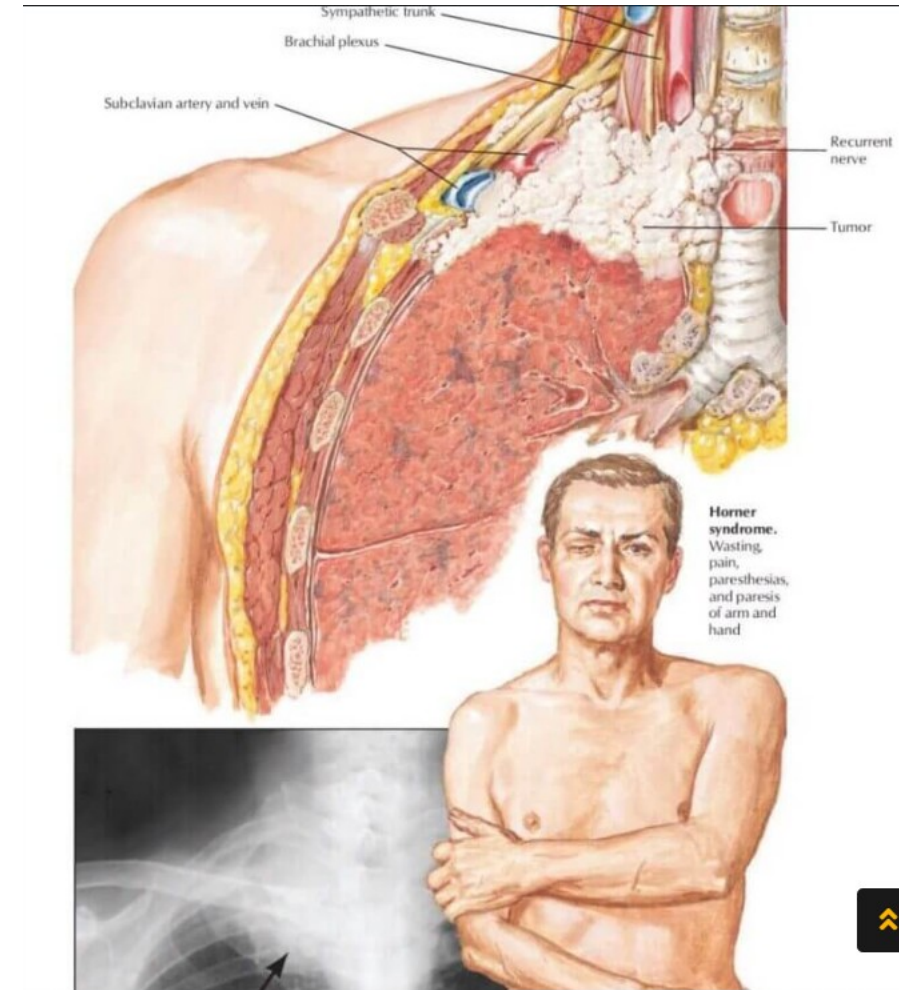
- ❖ **Definition:** A **diagnosis of exclusion**.
- ❖ A poorly differentiated NSCLC
- ❖ Lacks light microscopic features of adenocarcinoma, squamous cell, or small cell carcinoma.
- ❖ **Histopathology:**
  - Sheets of large, polygonal cells with prominent nucleoli and abundant cytoplasm.
  - No specific differentiation.





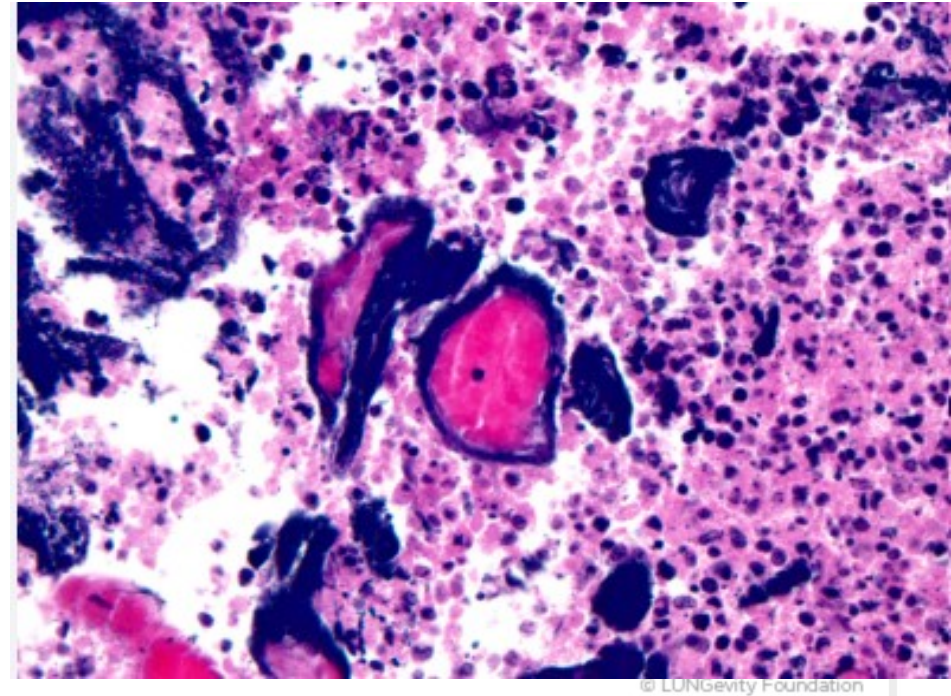
# Pancoast Tumor

- ❖ Also known as a **superior sulcus tumor**
- ❖ Is a **carcinoma located in the lung's apex** in the **superior pulmonary sulcus**.
- ❖ Due to its **location**, it readily invades nearby structures, leading to a characteristic constellation of symptoms known as **Pancoast Syndrome**.
- ❖ **Histology:** Most commonly (over 95%) are **non-small cell lung carcinomas (NSCLC)**.
- ❖ **The Classic Triad of Symptoms:**
  - Severe, Unrelenting Shoulder and Arm Pain
  - Horner's Syndrome
  - Muscular Atrophy of the Hand Muscles:



## C. Small Cell Lung Carcinoma (SCLC) (~15%)

- ❖ **Origin:** Neuroendocrine cells in the bronchial epithelium.
- ❖ **Location:** Almost always **central**.
- ❖ **Clinical: Highly aggressive, early metastases,** paraneoplastic syndromes common.
- ❖ Treated primarily with chemotherapy/radiation (rarely surgery).
- ❖ **Histopathology:**
  - Sheets of small, round, blue cells with:
    - "Salt-and-pepper" chromatin
    - High N:C ratio
    - Nuclear molding
    - Frequent mitoses and necrosis
    - Crush artifact (in small biopsies)



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## Paraneoplastic Syndromes

❖ **Systemic** manifestations not caused by the tumor mass or metastases.

### SCLC:

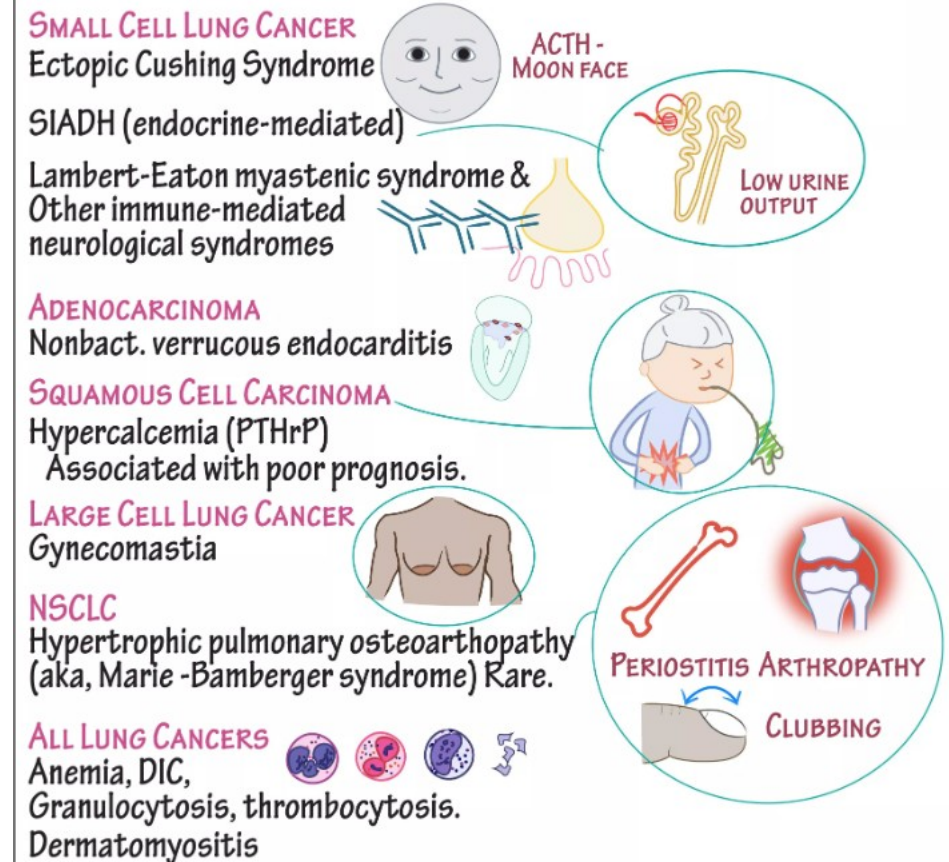
- ❖ **SIADH** (inappropriate ADH secretion)
- ❖ **Lambert-Eaton Myasthenic Syndrome**
- ❖ **Cushing's Syndrome** (ACTH production)

### Squamous Cell Carcinoma:

- ❖ **Hypercalcemia** (PTHrP production)

### Adenocarcinoma:

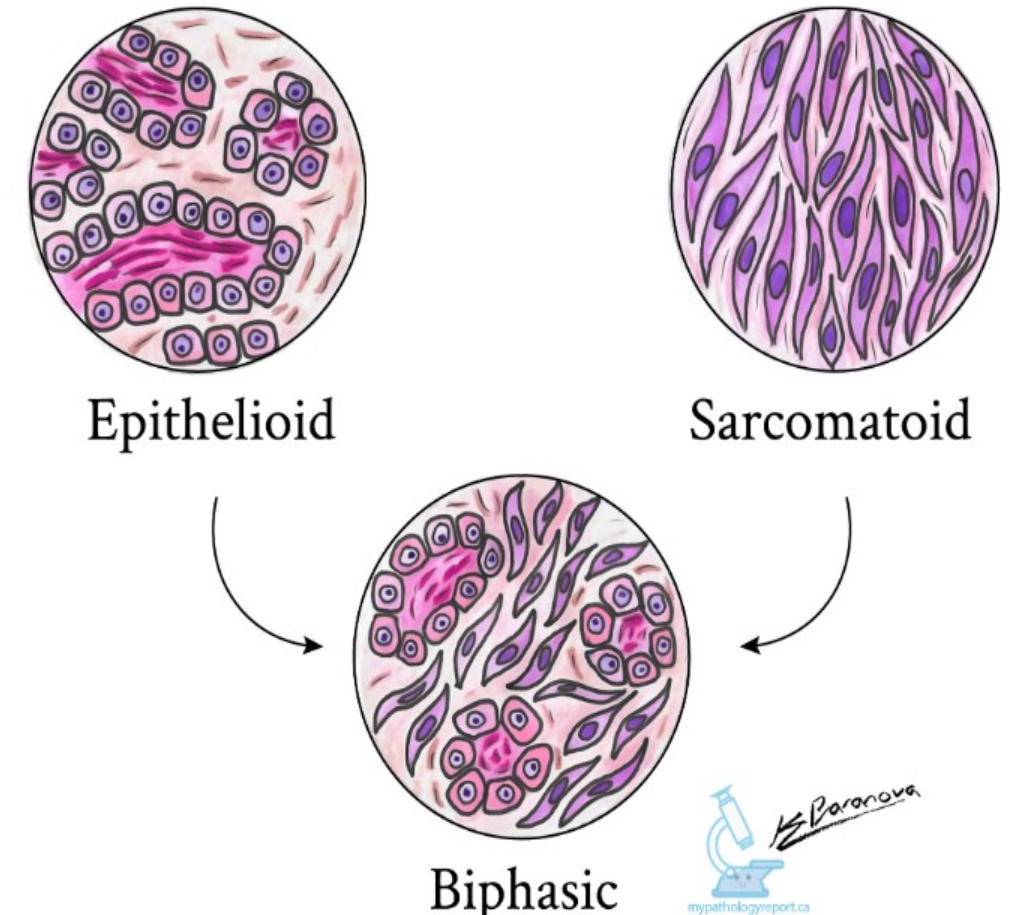
- ❖ **Hypertrophic Pulmonary Osteoarthropathy (HPOA)** and **digital clubbing**
- ❖ **Migratory Thrombophlebitis** (Trousseau syndrome)





# Mesothelioma

- ❖ **Origin:** Mesothelial cells of the pleura (rarely peritoneum).
- ❖ **Strongest Association: Asbestos exposure'**
- ❖ **Gross:** Diffuse, encasing growth of the lung ("pleural rind").
- ❖ **Histopathology:**
  - **Epithelioid** (most common): Tubulopapillary structures.
  - **Sarcomatoid:** Spindle cells.
  - **Biphasic:** Mixed.





# Metastatic Tumors to the Lung

❖ **Common Primary Sites:** Breast, Colon, Kidney, Melanoma, Sarcomas.

❖ **Pathology:**

- Often multiple, well-circumscribed, spherical nodules ("cannonball" metastases on imaging).
- Histology resembles the primary tumor.

